

Canonical GHZ configurations - cluster P4 - helicity conserving

species: electron
polarization cluster: P4
source cluster configurations: 81
canonical configurations collected: 15

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

helicity-mechanism subset: helicity conserving
classification: complete-chain R_flip/conserves ≤ 1
per-page labels: isolated flip/conserves and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:

- 2 components: 54 configurations
- 4 components: 8 configurations
- 5 components: 3 configurations
- 6 components: 14 configurations
- 7 components: 2 configurations

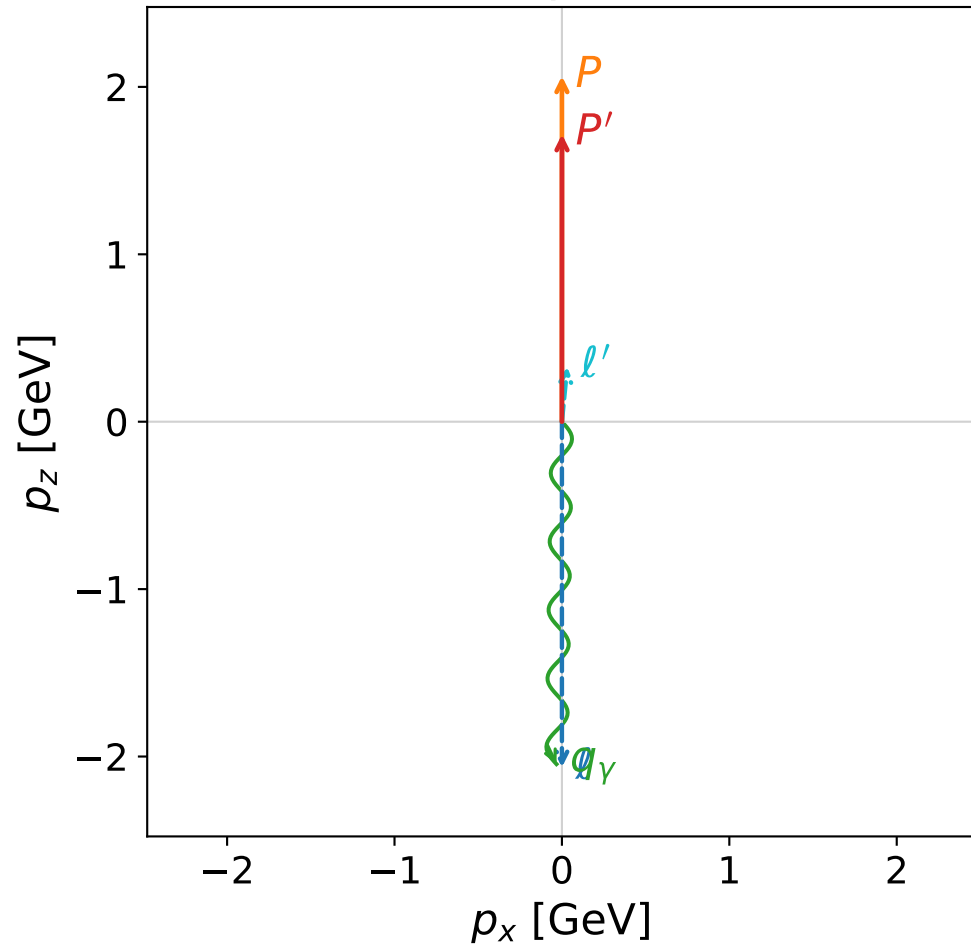
CSV index:

Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_04\combined\
helicity_conserving\canonical_dghz_polarization_cluster_04_helicity_conserving_configu
rations_electron.csv

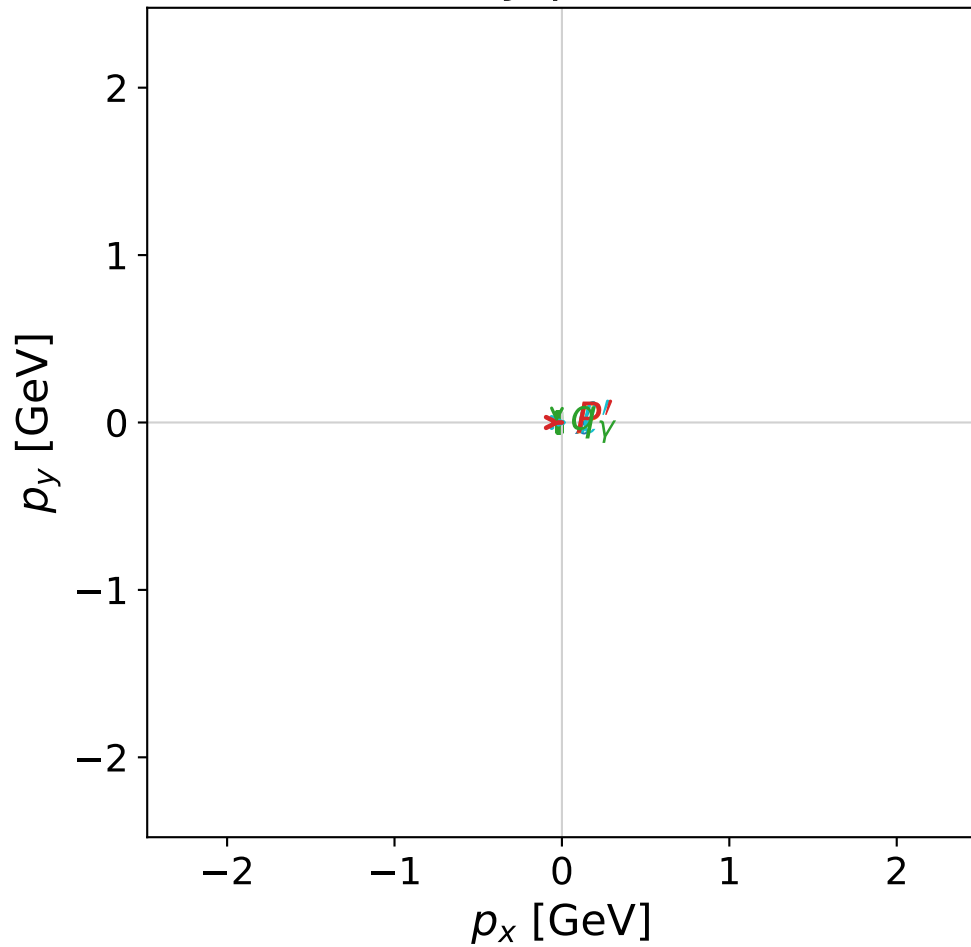
Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

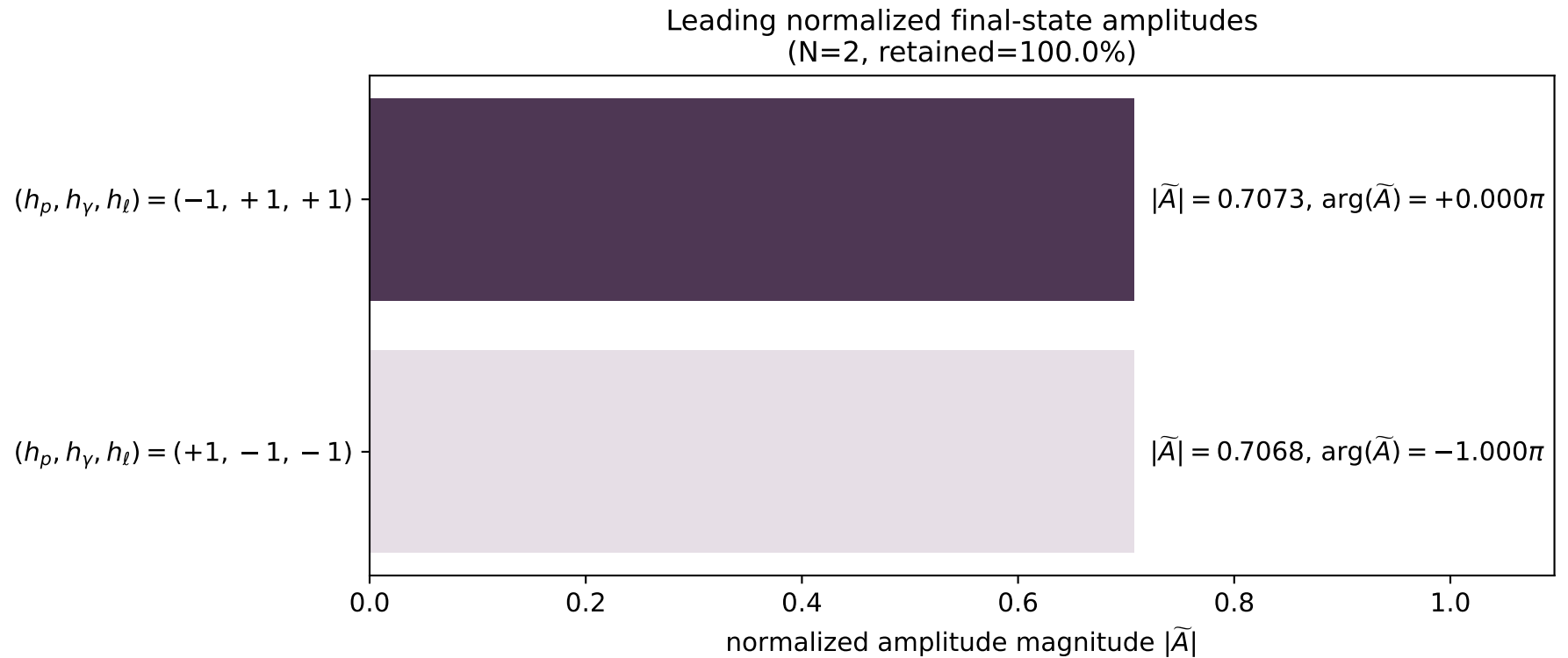


dghz_mixing_angles_polarization_04_local_minimum_184
 $d_{\{\rm GHZ\}}=2.62168\text{e-}06$
 region: polarization_cluster_04_local_minimum_184
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0184
 lepton mixing angle: $\alpha_e=2.1744032$ rad
 incoming lepton: $-0.567616|+\rangle +0.823294|-\rangle$
 proton mixing angle: $\alpha_p=2.5444445$ rad
 incoming proton: $-0.826943|+\rangle +0.562286|-\rangle$

$s=18.765$, $\sqrt{s}=4.33186$, $|\vec{P}|=2.06438$, $|\vec{P}'|=1.71704$
 $\theta_{p'}=0.000299242$, $\theta_\gamma=3.12573$, $\phi_{p'}=8.42046\text{e-}05$, $\phi_\gamma=3.14159$
 $E_\gamma=2.04553$, $\theta_{l'}=0.0969549$, $\phi_{l'}=6.28318$
 $Q^2=2.71682$, $x_B=0.153105$, $t=-0.023959$, $W^2=15.9079$, $y=0.992152$

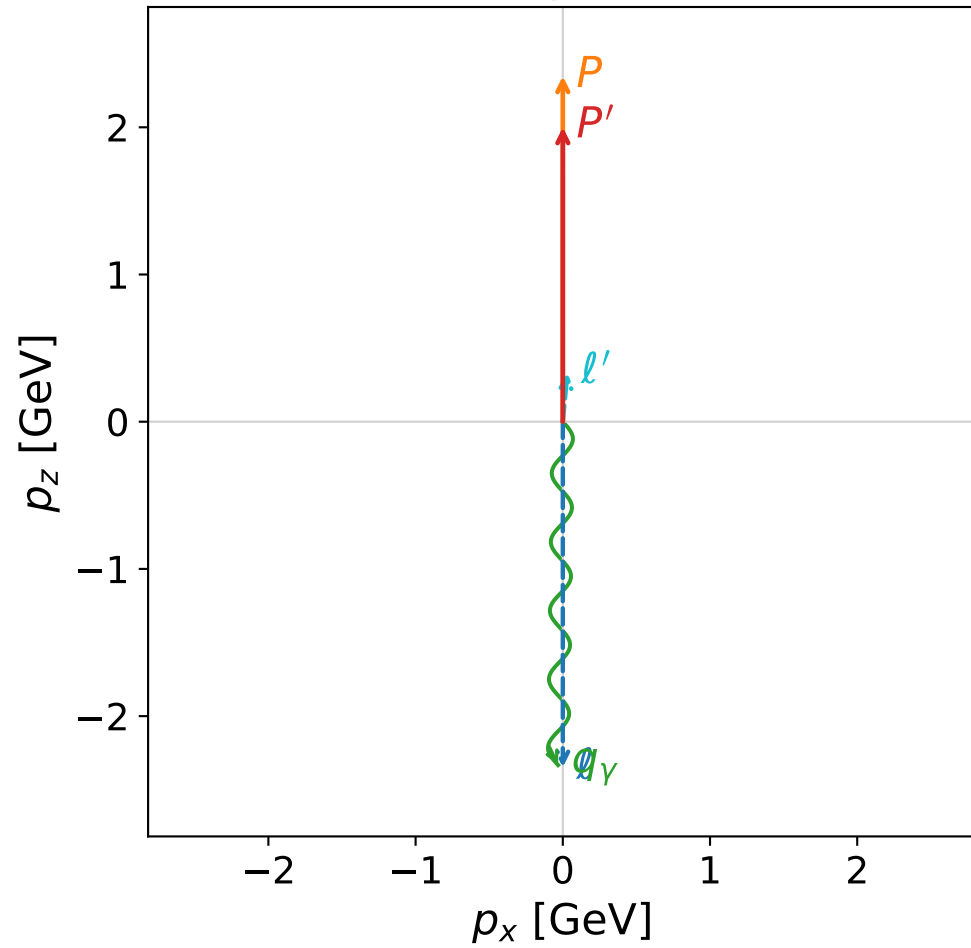
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0644$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0644$
 P $E= 2.2675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0644$
 l' $E= 0.3298$ $p_x= 0.0319$ $p_y= -0.0000$ $p_z= 0.3282$
 P' $E= 1.9565$ $p_x= 0.0005$ $p_y= 0.0000$ $p_z= 1.7170$
 q_γ $E= 2.0455$ $p_x= -0.0324$ $p_y= 0.0000$ $p_z= -2.0453$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.031%, conserve 99.969%
 $R_{\text{flip/conserve}}=3.059\text{e-}04$; $I_h=-9.057\text{e-}05$
 virtual-photon powers: T 99.985%, L 0.015%
 $R_{L/T}=1.475\text{e-}04$; $I_{TL}=+5.471\text{e-}04$

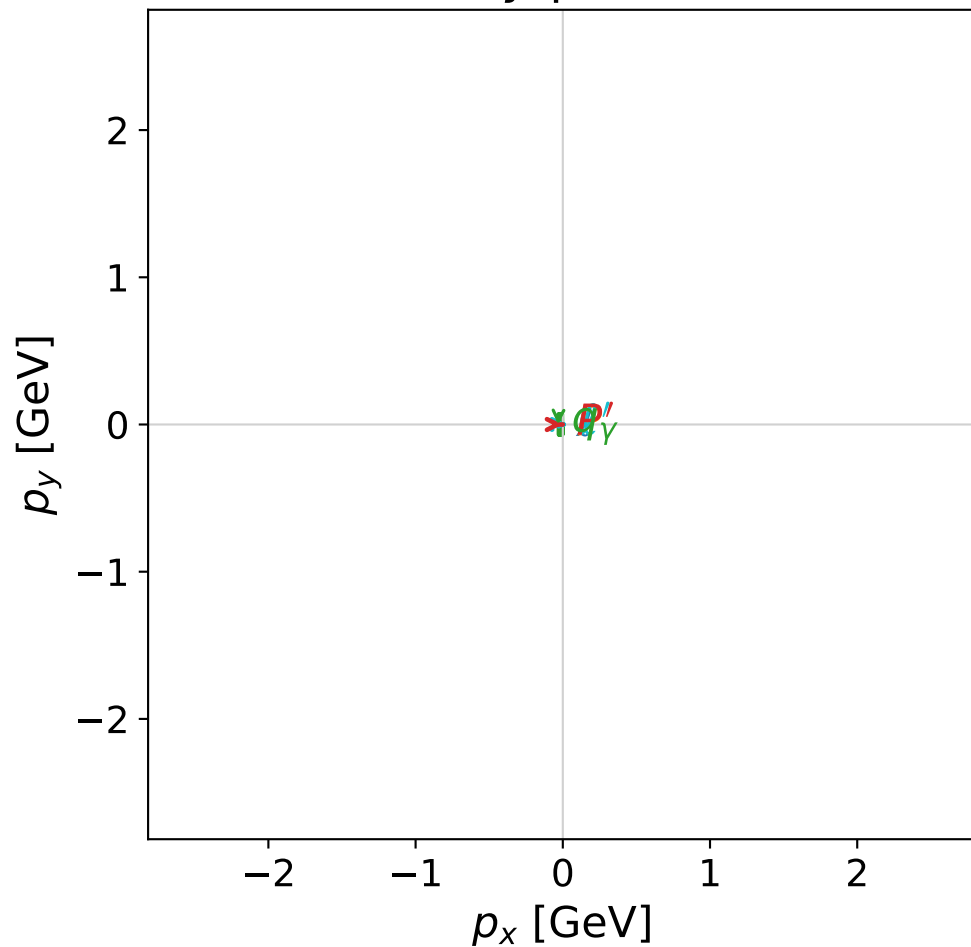


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

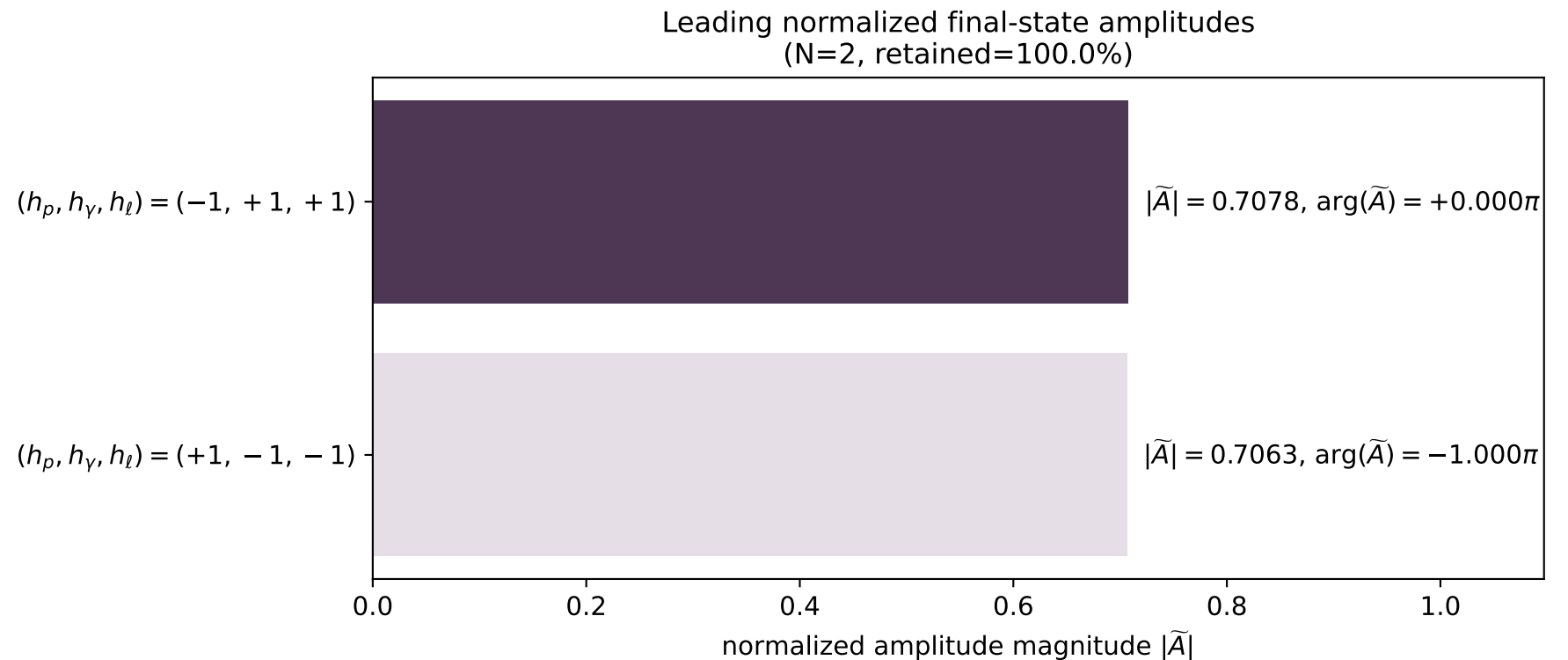


dghz_mixing_angles_polarization_04_local_minimum_186
 $d_{\text{GHZ}}=7.14153\text{e-}06$
 region: polarization_cluster_04_local_minimum_186
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0186
 lepton mixing angle: $\alpha_e=2.2229182$ rad
 incoming lepton: $-0.606874|+\rangle +0.794798|-\rangle$
 proton mixing angle: $\alpha_p=2.4957371$ rad
 incoming proton: $-0.798585|+\rangle +0.601882|-\rangle$

$s=23.7724$, $\sqrt{s}=4.8757$, $|\vec{P}|=2.34762$, $|\vec{P}'|=2.00059$
 $\theta_{p'}=0.000172458$, $\theta_\gamma=3.12791$, $\phi_{p'}=0.000436125$, $\phi_\gamma=3.14159$
 $E_\gamma=2.33272$, $\theta_{l'}=0.0948293$, $\phi_{l'}=6.28318$
 $Q^2=3.12386$, $x_B=0.137221$, $t=-0.0189857$, $W^2=20.5212$, $y=0.994436$

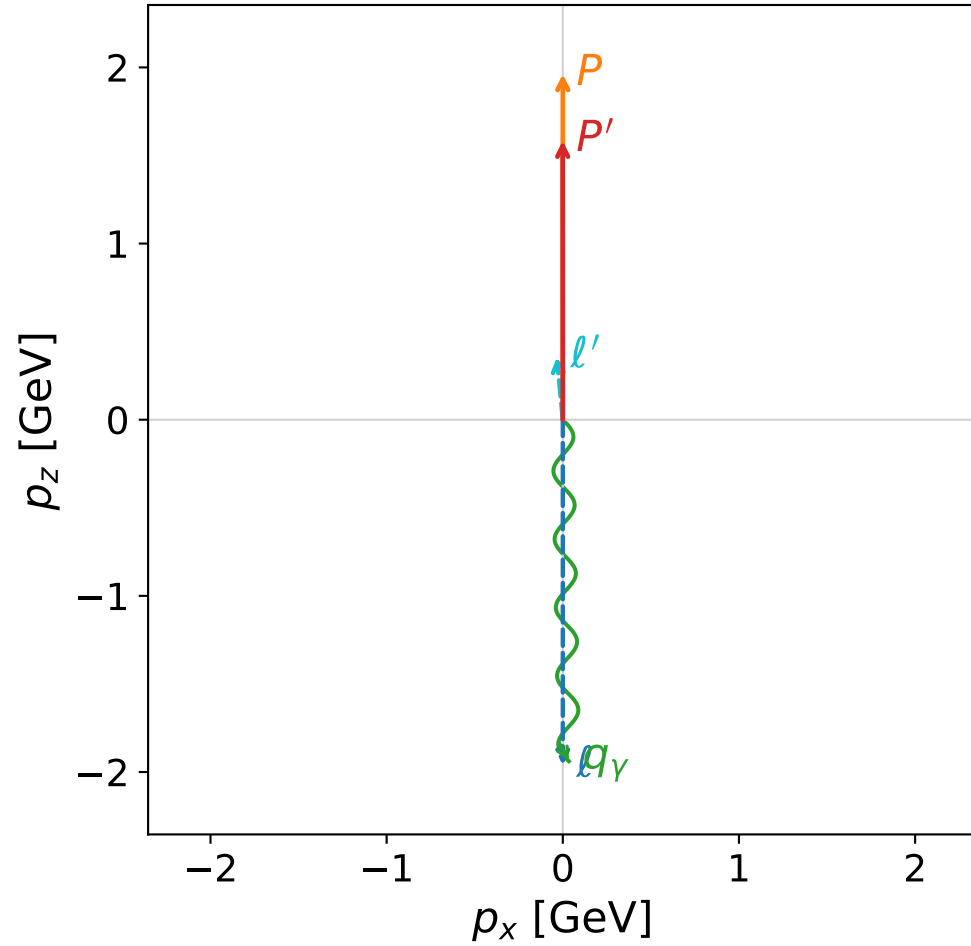
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3476$
 P $E= 2.5281$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3476$
 l' $E= 0.3334$ $p_x= 0.0316$ $p_y= -0.0000$ $p_z= 0.3319$
 P' $E= 2.2096$ $p_x= 0.0003$ $p_y= 0.0000$ $p_z= 2.0006$
 q_γ $E= 2.3327$ $p_x= -0.0319$ $p_y= 0.0000$ $p_z= -2.3325$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.029%, conserve 99.971%
 $R_{\text{flip/conserve}}=2.857\text{e-}04$; $I_h=-4.477\text{e-}05$
 virtual-photon powers: T 99.990%, L 0.010%
 $R_L/T=1.012\text{e-}04$; $I_{TL}=+2.666\text{e-}04$

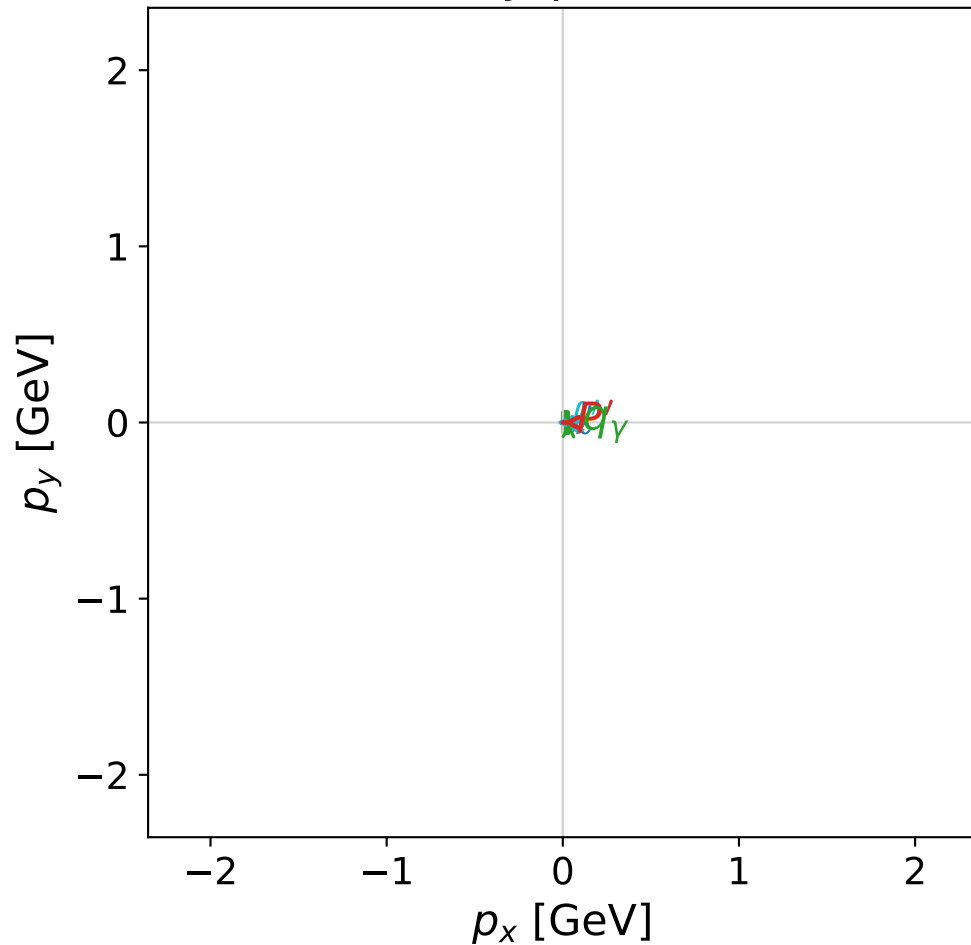


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_205

$d_{\text{GHZ}}=2.5832\text{e-}05$

region: polarization_cluster_04_local_minimum_205

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0205

lepton mixing angle: $\alpha_e=2.5877528$ rad

incoming lepton: $-0.850511|+\rangle +0.525957|-\rangle$

proton mixing angle: $\alpha_p=2.116265$ rad

incoming proton: $-0.518819|+\rangle +0.854884|-\rangle$

$s=17.1102$, $\sqrt{s}=4.13645$, $|\vec{P}|=1.96187$, $|\vec{P}'|=1.58344$

$\theta_{P'}=0.000451651$, $\theta_\gamma=3.1237$, $\phi_{P'}=3.14162$, $\phi_\gamma=6.28317$

$E_\gamma=1.93908$, $\theta_{l'}=0.0953256$, $\phi_{l'}=3.14157$

$Q^2=2.79482$, $x_B=0.173893$, $t=-0.0315471$, $W^2=14.1571$, $y=0.990252$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 1.9619$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9619$

P $E= 2.1746$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9619$

l' $E= 0.3570$ $p_x= -0.0340$ $p_y= 0.0000$ $p_z= 0.3553$

P' $E= 1.8404$ $p_x= -0.0007$ $p_y= -0.0000$ $p_z= 1.5834$

q_γ $E= 1.9391$ $p_x= 0.0347$ $p_y= -0.0000$ $p_z= -1.9388$

GHZ mechanism: helicity conserving (full H G C)

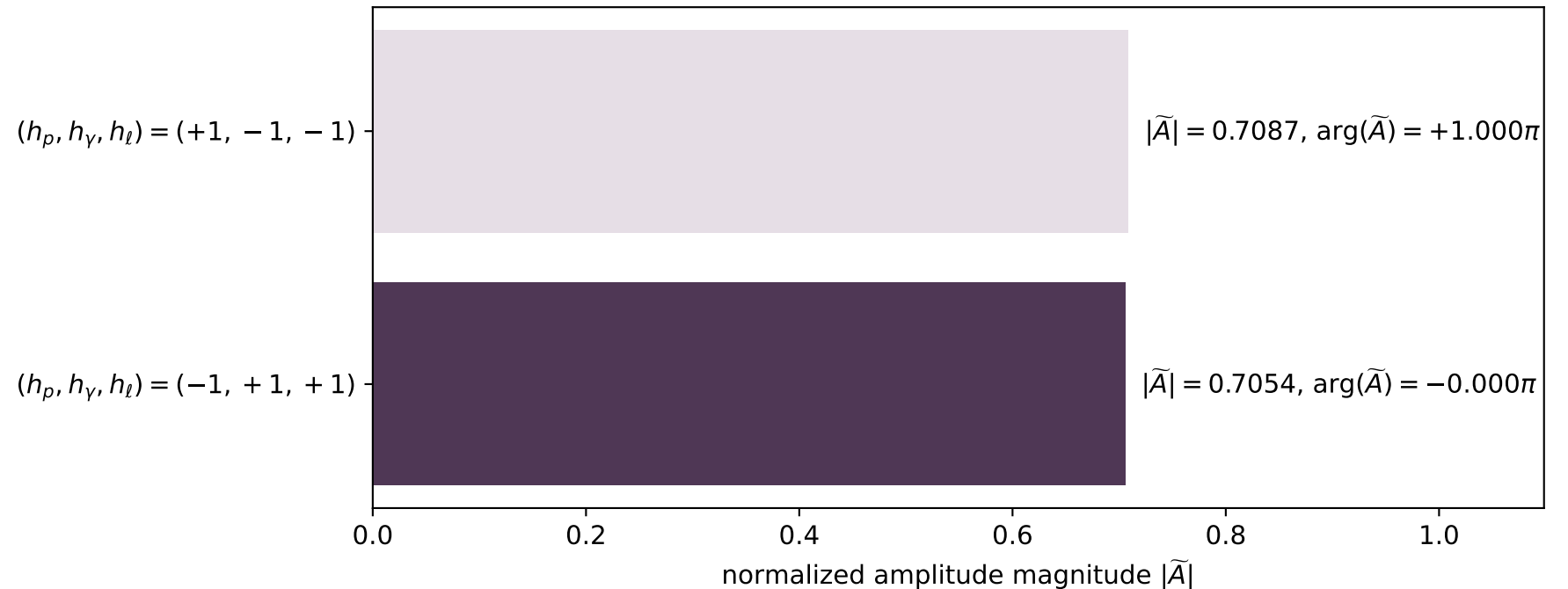
helicity powers: flip 0.031%, conserve 99.969%

$R_{\text{flip/conserve}}=3.081\text{e-}04$; $I_h=-1.465\text{e-}04$

virtual-photon powers: T 99.983%, L 0.017%

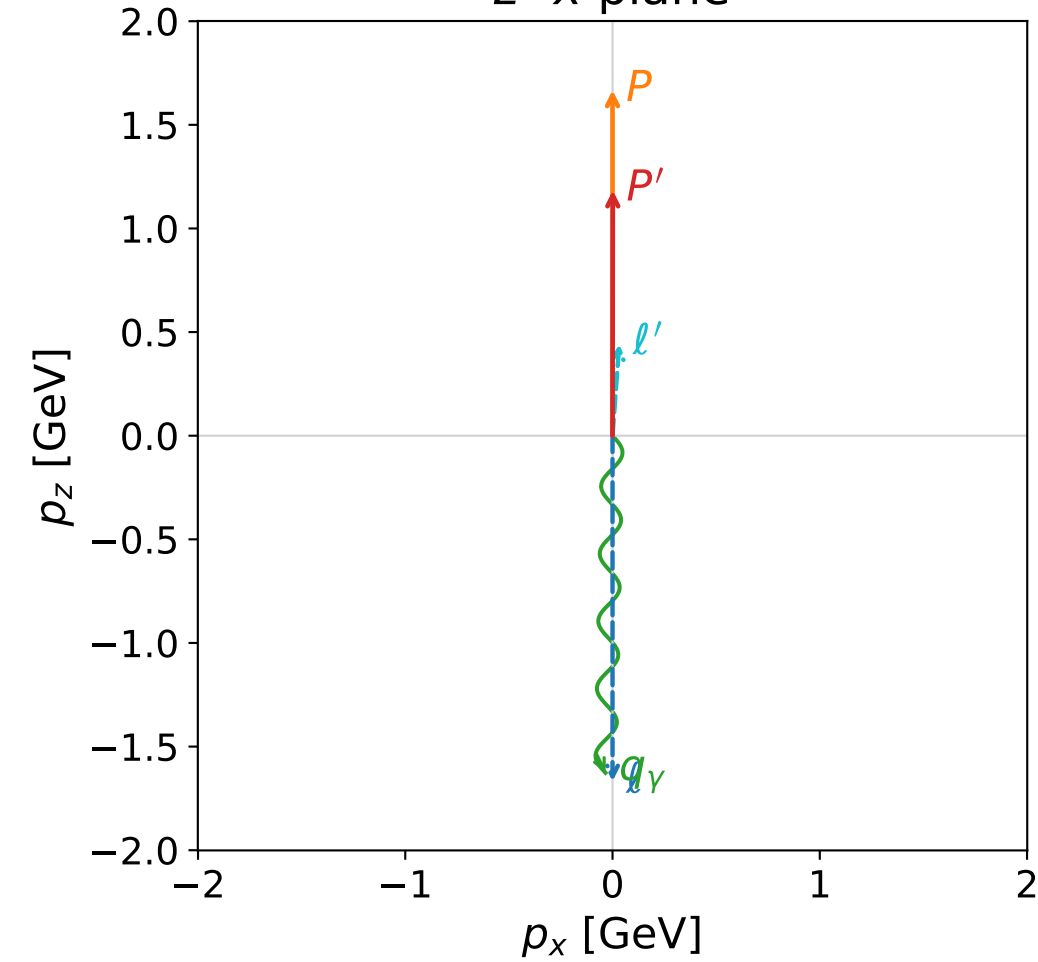
$R_{\text{L/T}}=1.745\text{e-}04$; $I_{\text{TL}}=+8.286\text{e-}04$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

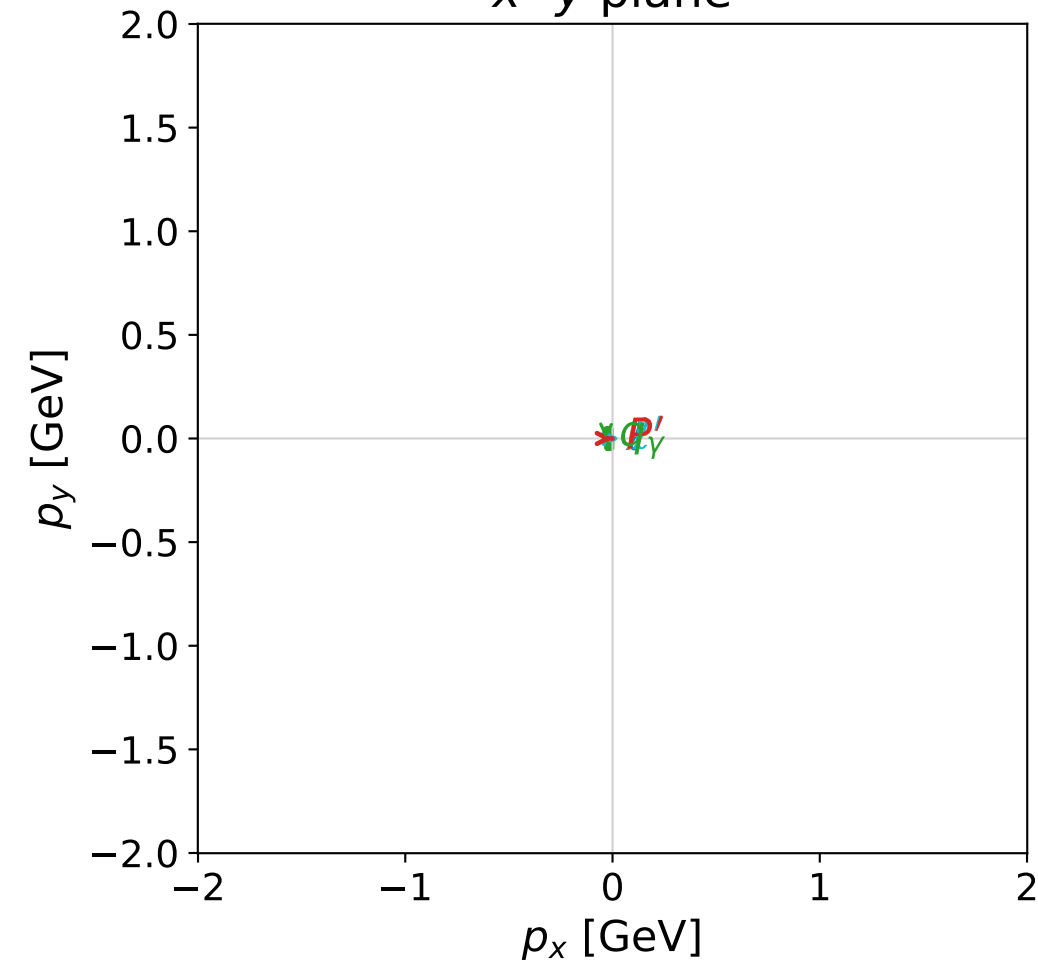


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



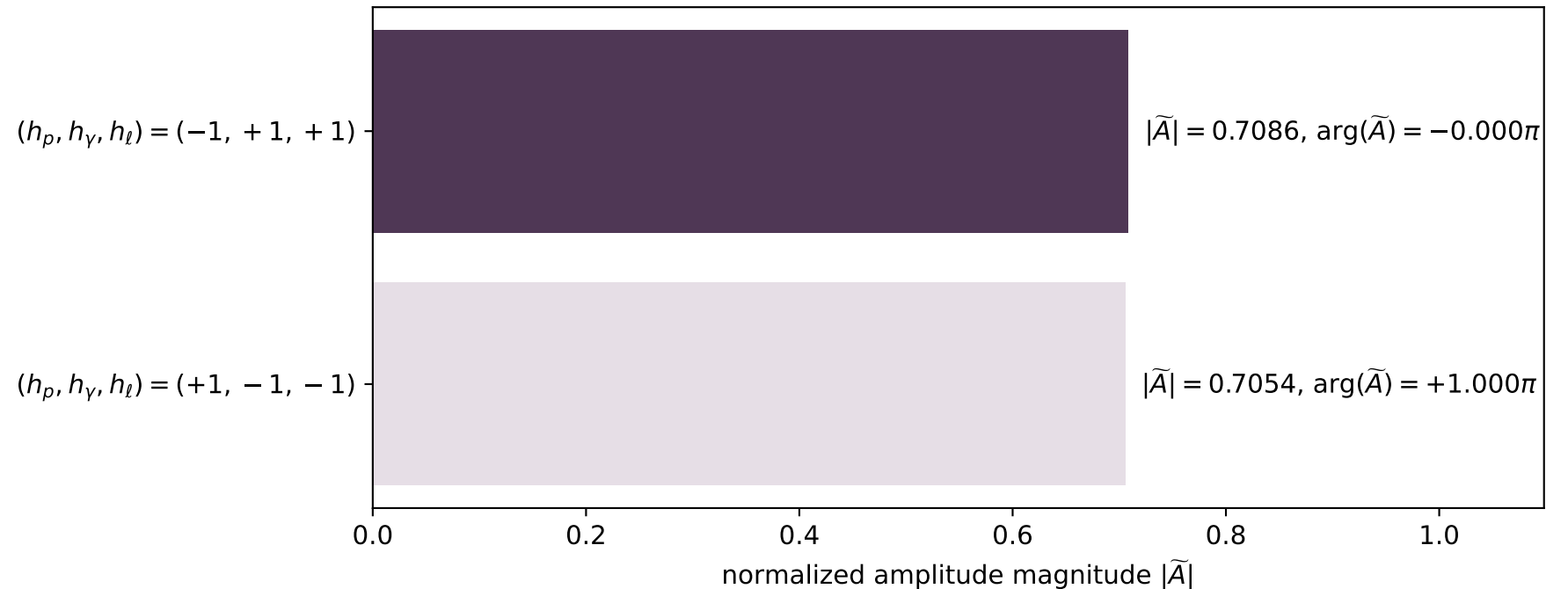
dghz_mixing_angles_polarization_04_local_minimum_210
 $d_{\text{GHZ}}=2.82327\text{e-}05$
 region: polarization_cluster_04_local_minimum_210
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0210
 lepton mixing angle: $\alpha_e=1.938937$ rad
 incoming lepton: $-0.359881|+\rangle +0.932998|-\rangle$
 proton mixing angle: $\alpha_p=2.7798744$ rad
 incoming proton: $-0.93529|+\rangle +0.353882|-\rangle$

$s=12.8213$, $\sqrt{s}=3.58068$, $|\vec{P}|=1.66748$, $|\vec{P}'|=1.1839$
 $\theta_{P'}=0.00120427$, $\theta_\gamma=3.12072$, $\phi_{P'}=6.28227$, $\phi_\gamma=3.14164$
 $E_\gamma=1.62664$, $\theta_{l'}=0.0733953$, $\phi_{l'}=8.56722\text{e-}05$
 $Q^2=2.95466$, $x_B=0.252116$, $t=-0.0716446$, $W^2=9.64462$, $y=0.981411$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.6675$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.6675$
 P $E= 1.9132$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.6675$
 l' $E= 0.4436$ $p_x= 0.0325$ $p_y= 0.0000$ $p_z= 0.4424$
 P' $E= 1.5105$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1839$
 q_γ $E= 1.6266$ $p_x= -0.0340$ $p_y= -0.0000$ $p_z= -1.6263$

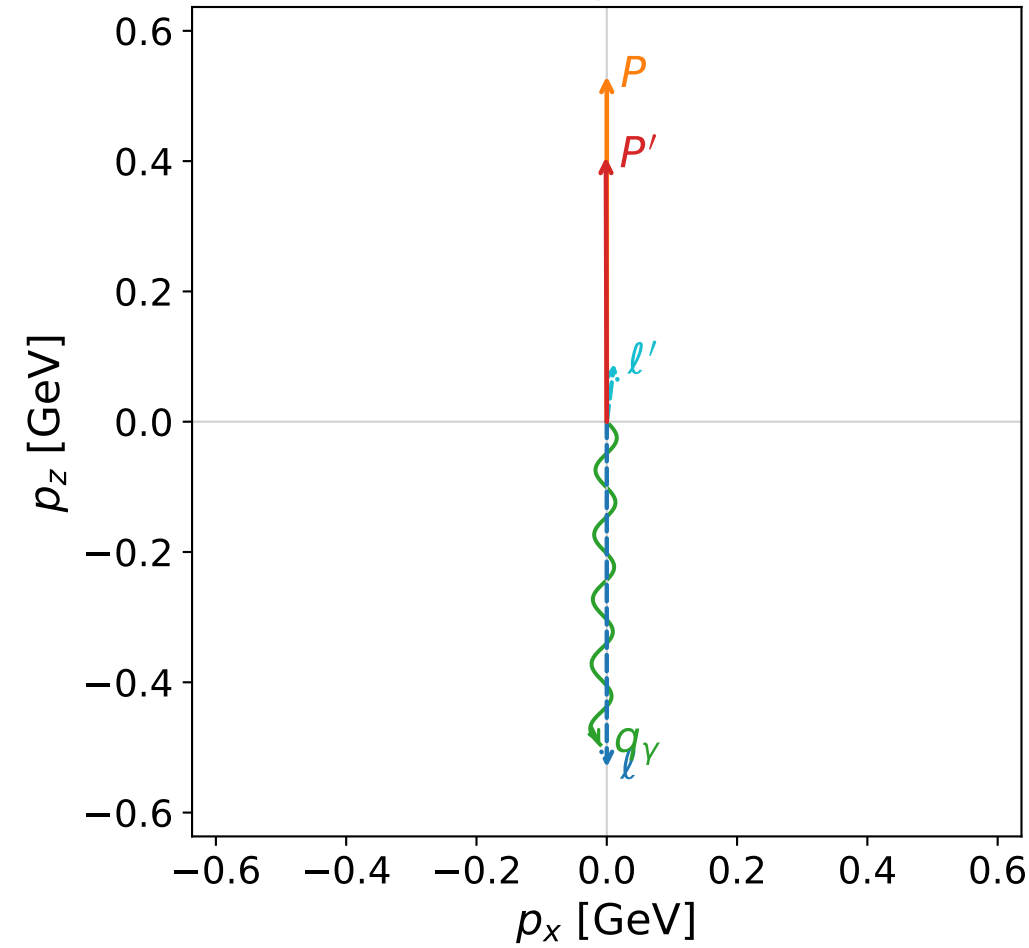
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.070%, conserve 99.930%
 $R_{\text{flip/conserve}}=7.003\text{e-}04$; $I_h=-7.342\text{e-}04$
 virtual-photon powers: T 99.973%, L 0.027%
 $R_L/T=2.667\text{e-}04$; $I_{TL}=+2.092\text{e-}03$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

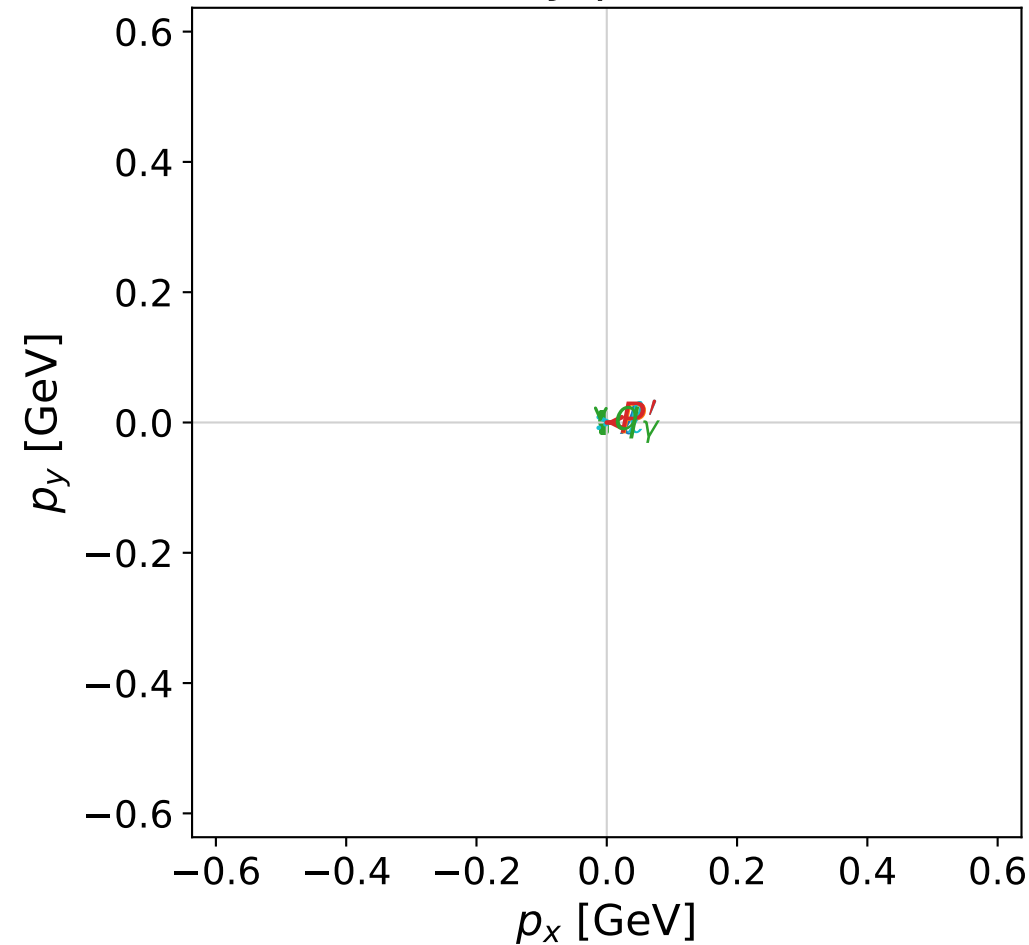


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

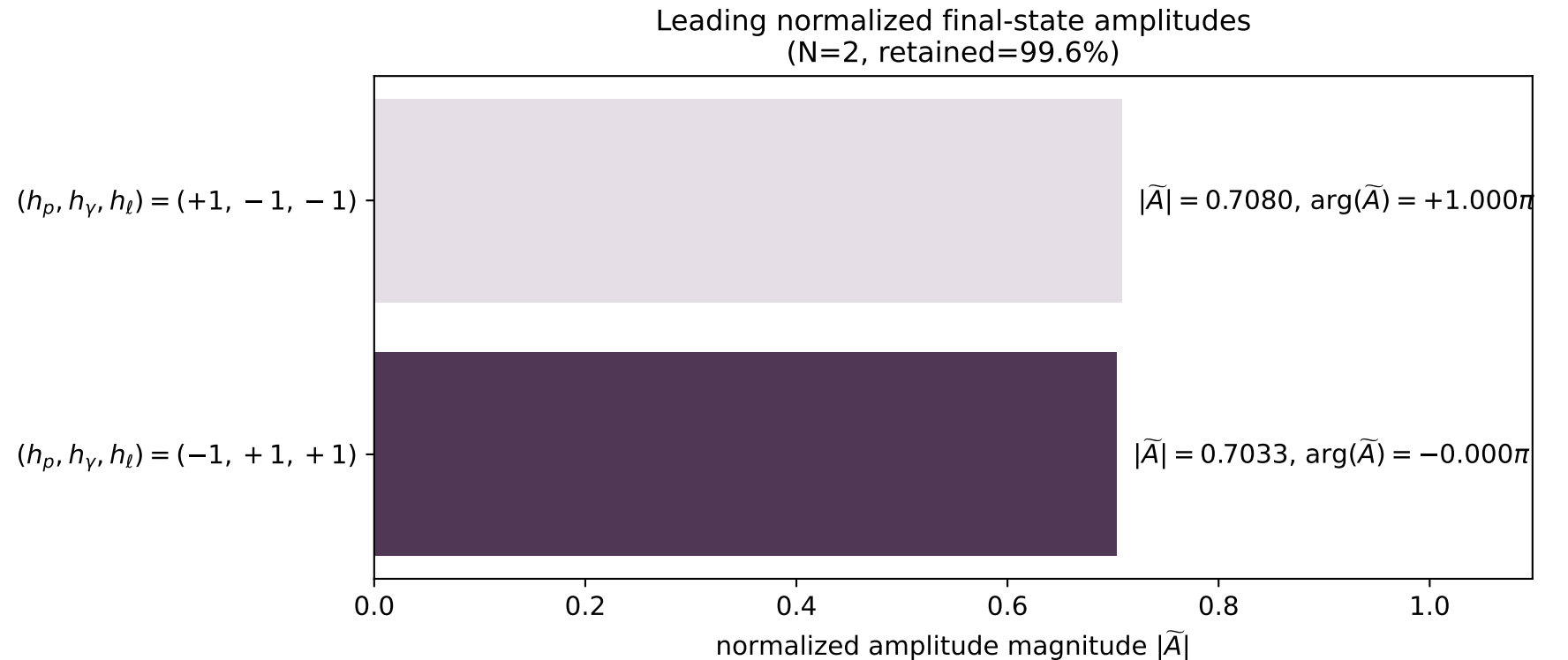


dghz_mixing_angles_polarization_04_local_minimum_221
 $d_{\text{GHZ}}=4.54196\text{e-}05$
 region: polarization_cluster_04_local_minimum_221
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0221
 lepton mixing angle: $\alpha_e=1.989916$ rad
 incoming lepton: $-0.406956|+\rangle +0.913448|-\rangle$
 proton mixing angle: $\alpha_p=2.6885858$ rad
 incoming proton: $-0.899135|+\rangle +0.437671|-\rangle$

$s=2.58602$, $\sqrt{s}=1.60811$, $|\vec{P}|=0.53049$, $|\vec{P}'|=0.406256$
 $\theta_{P'}=0.00318369$, $\theta_\gamma=3.12141$, $\phi_{P'}=3.14161$, $\phi_\gamma=3.14159$
 $E_\gamma=0.495777$, $\theta_{l'}=0.125661$, $\phi_{l'}=6.28318$
 $Q^2=0.190506$, $x_B=0.118564$, $t=-0.0123647$, $W^2=2.29612$, $y=0.941749$

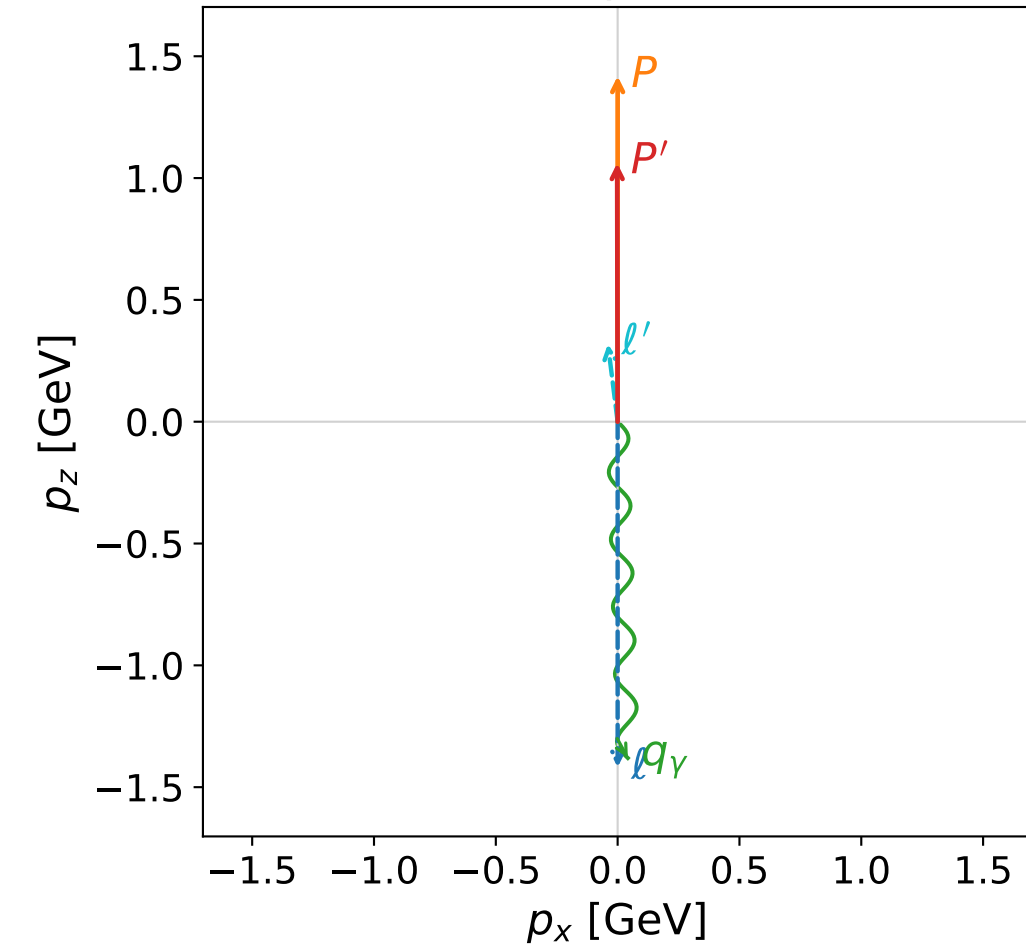
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.5305$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5305$
 P $E= 1.0776$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5305$
 l' $E= 0.0901$ $p_x= 0.0113$ $p_y= -0.0000$ $p_z= 0.0894$
 P' $E= 1.0222$ $p_x= -0.0013$ $p_y= -0.0000$ $p_z= 0.4063$
 q_γ $E= 0.4958$ $p_x= -0.0100$ $p_y= 0.0000$ $p_z= -0.4957$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.485%, conserve 99.515%
 $R_{\text{flip/conserve}}=4.872\text{e-}03$; $I_h=-4.113\text{e-}03$
 virtual-photon powers: T 99.410%, L 0.590%
 $R_L/T=5.938\text{e-}03$; $I_{TL}=-3.661\text{e-}02$

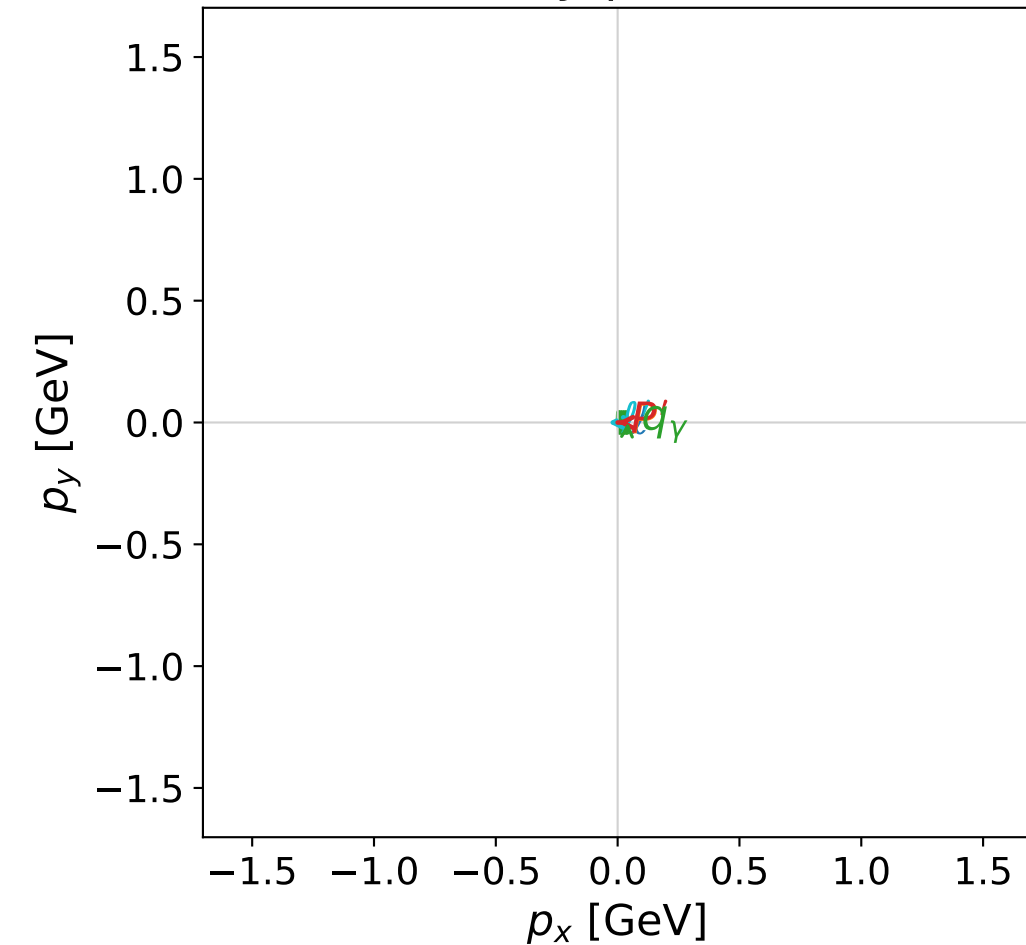


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

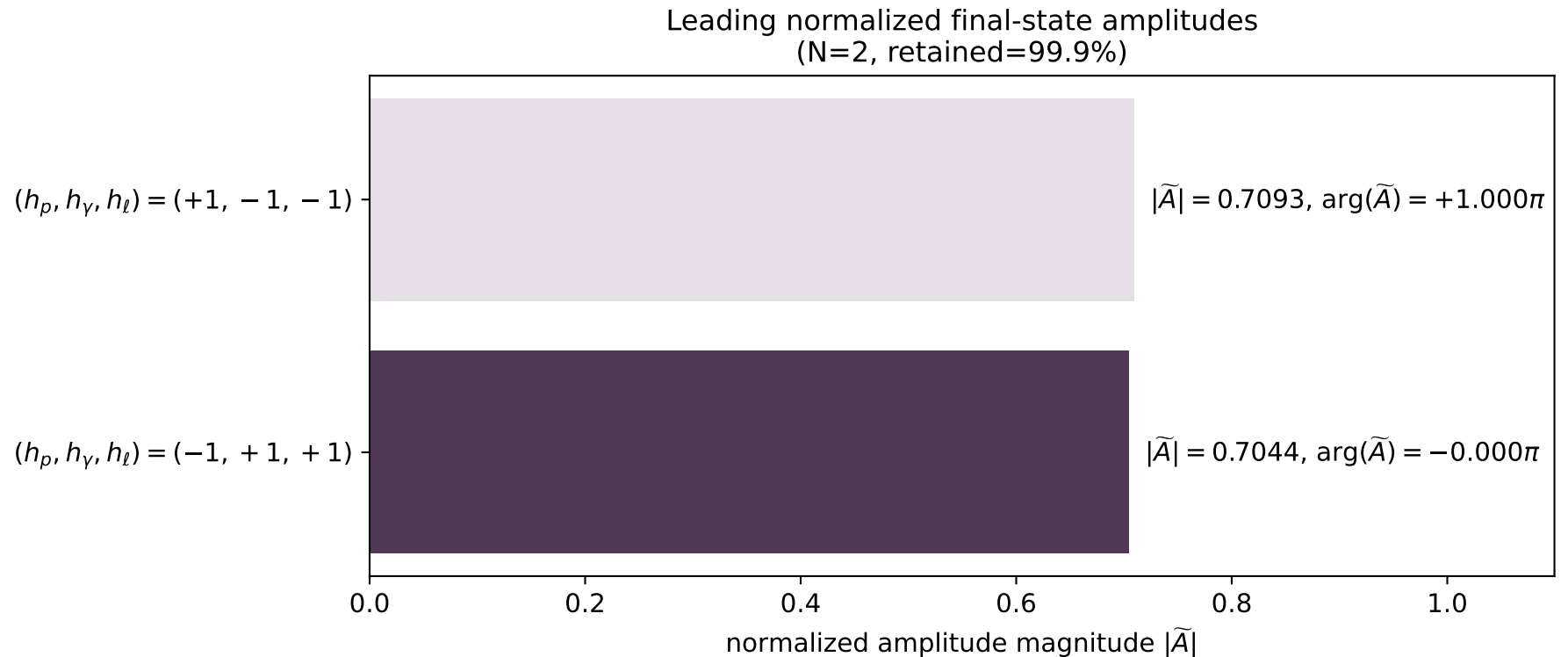


dghz_mixing_angles_polarization_04_local_minimum_228
 $d_{\text{GHZ}}=5.43438\text{e-}05$
 region: polarization_cluster_04_local_minimum_228
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0228
 lepton mixing angle: $\alpha_e=2.7339948$ rad
 incoming lepton: $-0.918076|+\rangle +0.396405|-\rangle$
 proton mixing angle: $\alpha_p=1.9650585$ rad
 incoming proton: $-0.384127|+\rangle +0.92328|-\rangle$

$s=9.72826$, $\sqrt{s}=3.11902$, $|\vec{P}|=1.41846$, $|\vec{P}'|=1.06154$
 $\theta_{P'}=0.00179591$, $\theta_\gamma=3.11125$, $\phi_{P'}=3.1417$, $\phi_\gamma=2.99541\text{e-}05$
 $E_\gamma=1.38105$, $\theta_{l'}=0.124754$, $\phi_{l'}=3.14162$
 $Q^2=1.81636$, $x_B=0.209741$, $t=-0.0467601$, $W^2=7.7235$, $y=0.978708$

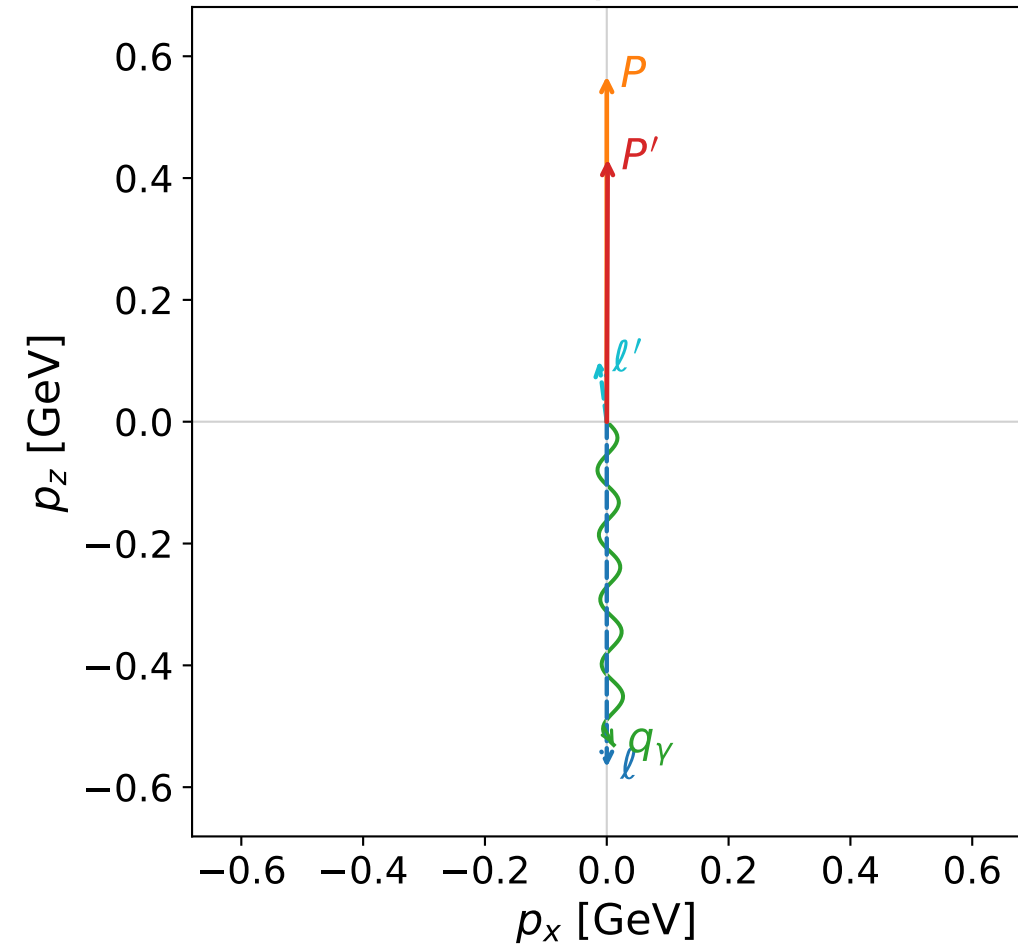
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.4185$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4185$
 P $E= 1.7006$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4185$
 l' $E= 0.3214$ $p_x= -0.0400$ $p_y= -0.0000$ $p_z= 0.3189$
 P' $E= 1.4166$ $p_x= -0.0019$ $p_y= -0.0000$ $p_z= 1.0615$
 q_γ $E= 1.3811$ $p_x= 0.0419$ $p_y= 0.0000$ $p_z= -1.3804$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.037%, conserve 99.963%
 $R_{\text{flip/conserve}}=3.661\text{e-}04$; $I_h=-5.049\text{e-}04$
 virtual-photon powers: T 99.907%, L 0.093%
 $R_{L/T}=9.266\text{e-}04$; $I_{TL}=+4.687\text{e-}03$

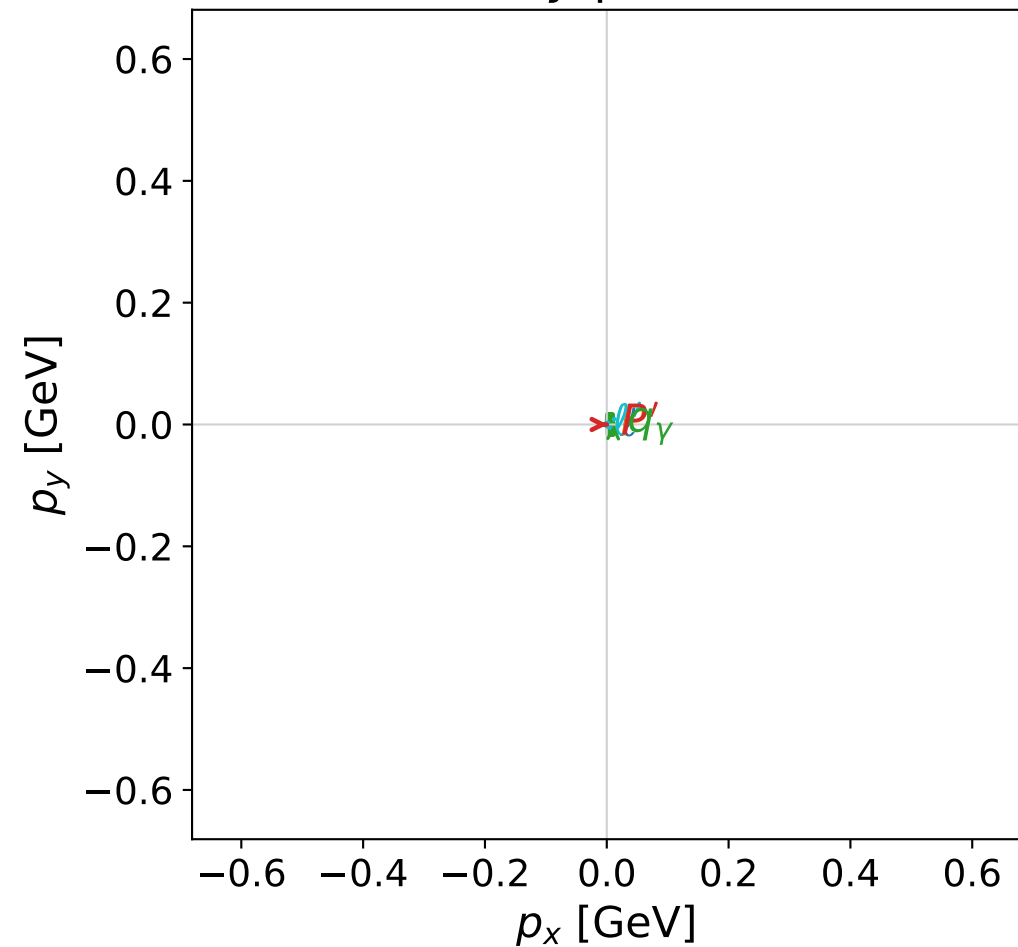


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

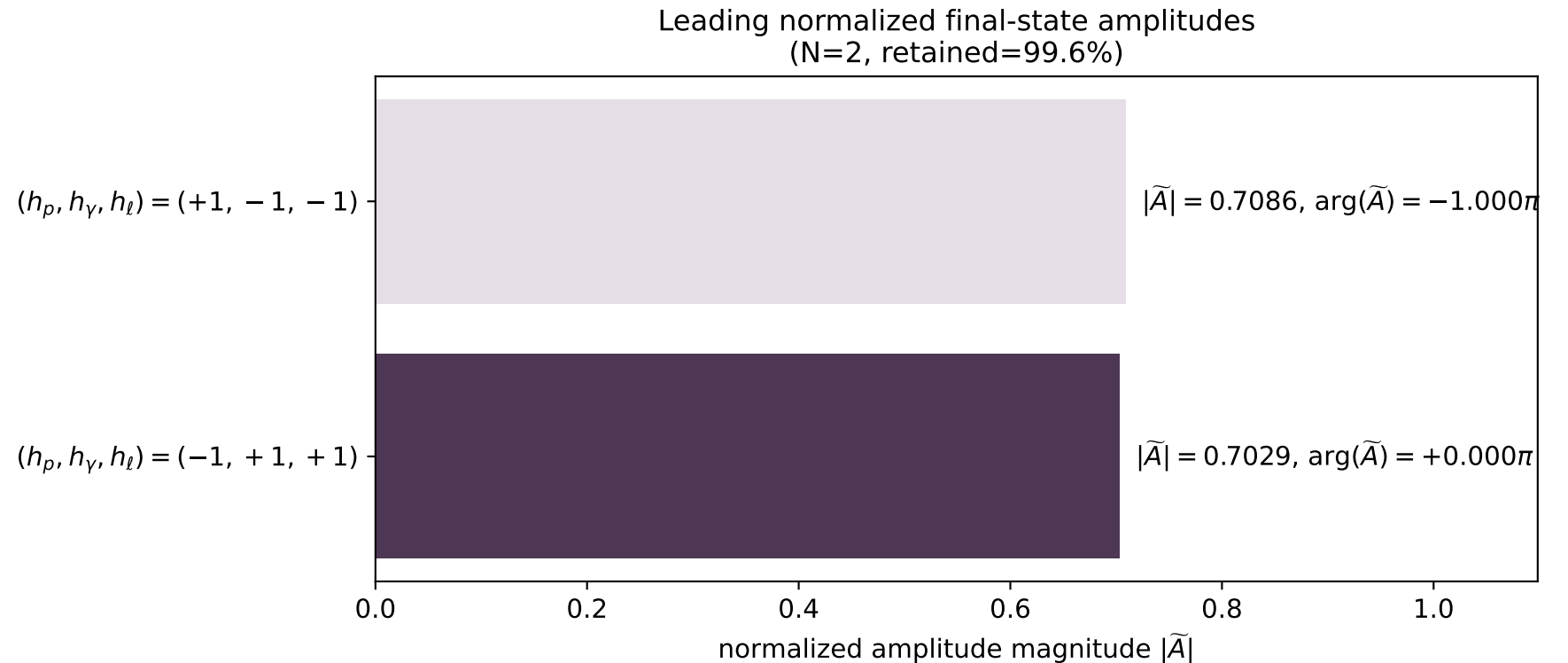


dghz_mixing_angles_polarization_04_local_minimum_235
 $d_{\{\rm GHZ\}}=6.62947\text{e-}05$
 region: polarization_cluster_04_local_minimum_235
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0235
 lepton mixing angle: $\alpha_e=2.7359514$ rad
 incoming lepton: $-0.91885|+\rangle +0.394608|-\rangle$
 proton mixing angle: $\alpha_p=2.0031027$ rad
 incoming proton: $-0.418966|+\rangle +0.908002|-\rangle$

$s=2.76767$, $\sqrt{s}=1.66363$, $|\vec{P}|=0.567381$, $|\vec{P}'|=0.431117$
 $\theta_{p'}=0.00334598$, $\theta_\gamma=3.12$, $\phi_{p'}=4.16802\text{e-}06$, $\phi_\gamma=6.28316$
 $E_\gamma=0.530853$, $\theta_{\ell'}=0.128826$, $\phi_{\ell'}=3.14157$
 $Q^2=0.227024$, $x_B=0.127496$, $t=-0.0144849$, $W^2=2.43345$, $y=0.943218$

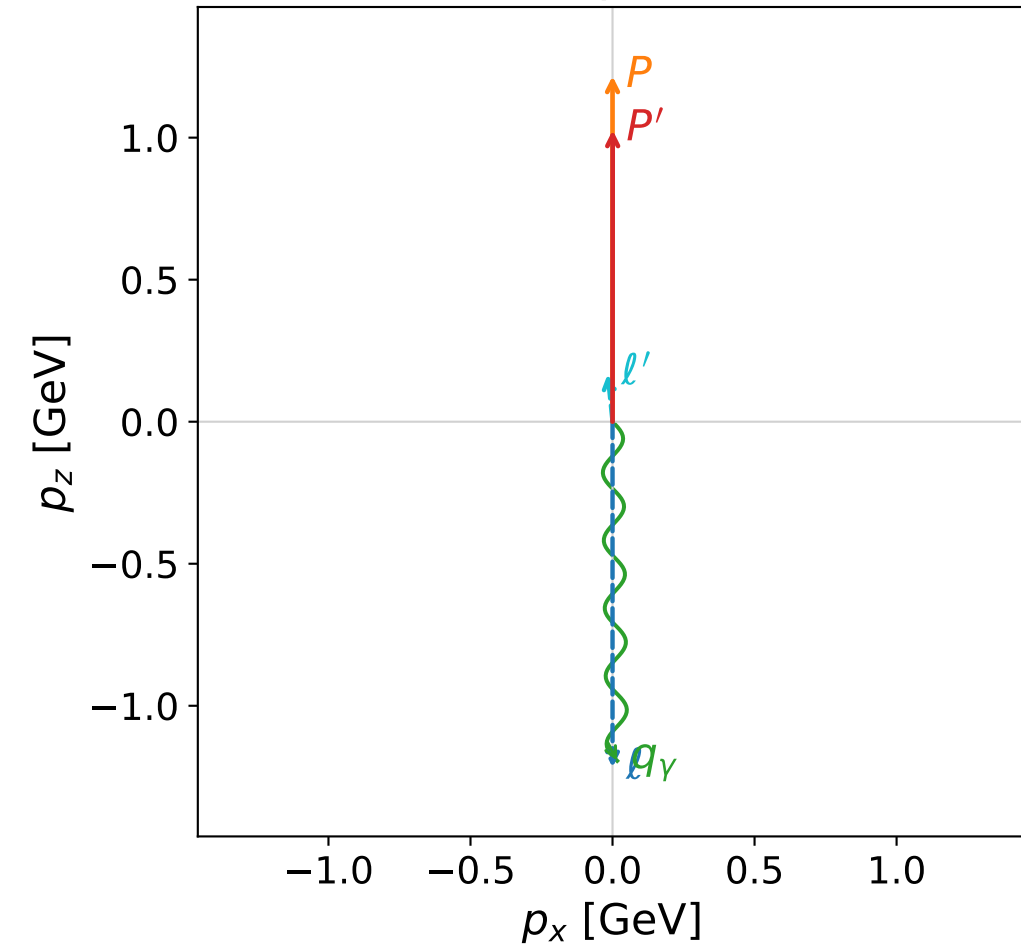
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.5674$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5674$
 P $E= 1.0963$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5674$
 ℓ' $E= 0.1004$ $p_x= -0.0129$ $p_y= 0.0000$ $p_z= 0.0996$
 P' $E= 1.0323$ $p_x= 0.0014$ $p_y= 0.0000$ $p_z= 0.4311$
 q_γ $E= 0.5309$ $p_x= 0.0115$ $p_y= -0.0000$ $p_z= -0.5307$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.404%, conserve 99.596%
 $R_{\text{flip/conserve}}=4.061\text{e-}03$; $I_h=-3.910\text{e-}03$
 virtual-photon powers: T 99.435%, L 0.565%
 $R_{L/T}=5.681\text{e-}03$; $I_{TL}=-3.555\text{e-}02$

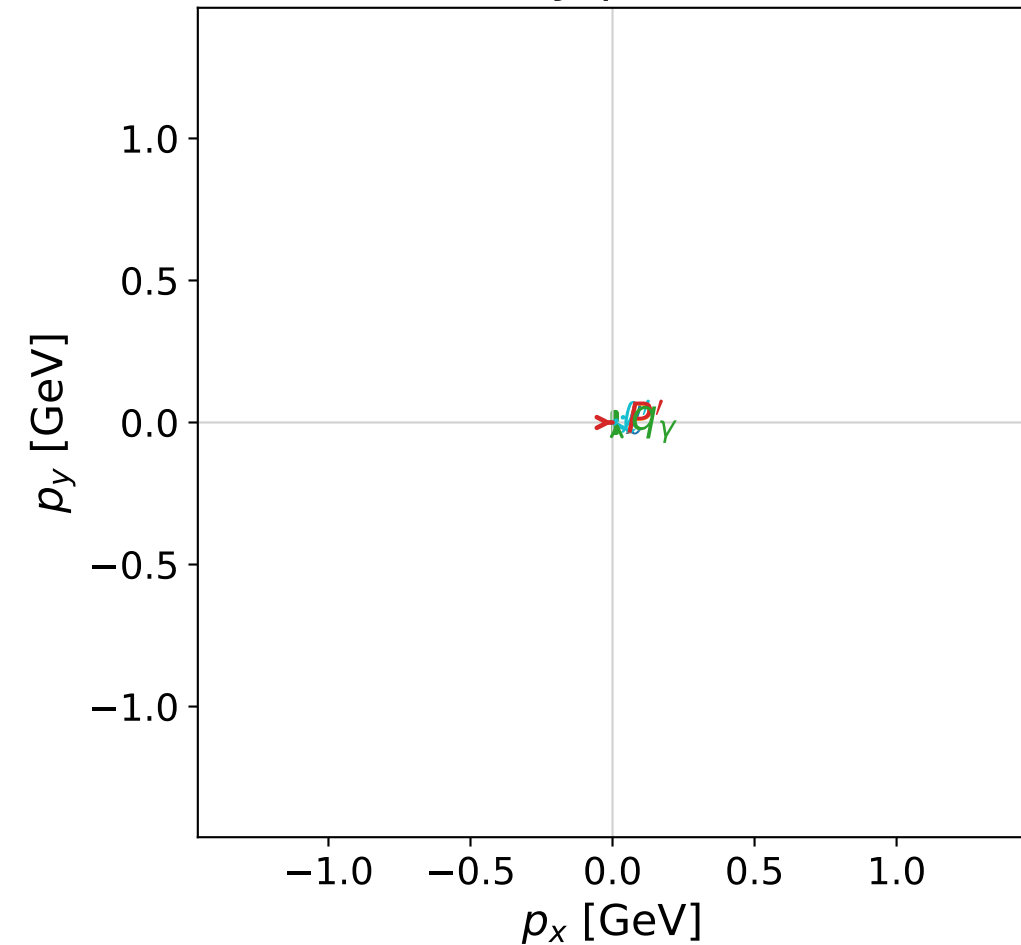


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_238

$d_{\{\rm GHZ\}}=7.24626\text{e-}05$

region: polarization_cluster_04_local_minimum_238

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0238

lepton mixing angle: $\alpha_e=2.5683975$ rad

incoming lepton: $-0.840172|+\rangle +0.542319|-\rangle$

proton mixing angle: $\alpha_p=2.1590186$ rad

incoming proton: $-0.554883|+\rangle +0.831928|-\rangle$

$s=7.57815$, $\sqrt{s}=2.75284$, $|\vec{P}|=1.21662$, $|\vec{P}'|=1.02635$

$\theta_{p'}=0.000568325$, $\theta_\gamma=3.12781$, $\phi_{p'}=0.000173411$, $\phi_\gamma=8.59629\text{e-}06$

$E_\gamma=1.19402$, $\theta_{\ell'}=0.101329$, $\phi_{\ell'}=3.14161$

$Q^2=0.817507$, $x_B=0.12408$, $t=-0.0149387$, $W^2=6.65088$, $y=0.983613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 1.2166$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2166$

P $E= 1.5362$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2166$

ℓ' $E= 0.1684$ $p_x= -0.0170$ $p_y= -0.0000$ $p_z= 0.1676$

P' $E= 1.3904$ $p_x= 0.0006$ $p_y= 0.0000$ $p_z= 1.0263$

q_γ $E= 1.1940$ $p_x= 0.0165$ $p_y= 0.0000$ $p_z= -1.1939$

GHZ mechanism: helicity conserving (full H G C)

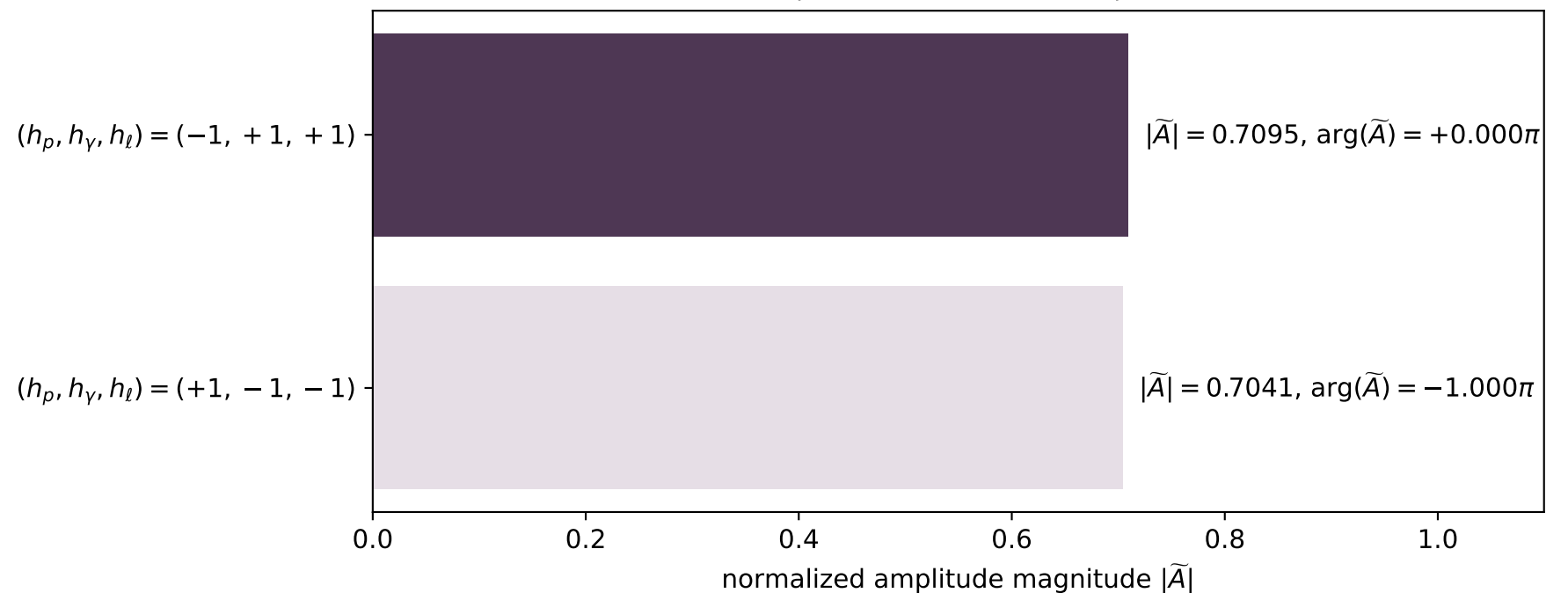
helicity powers: flip 0.123%, conserve 99.877%

$R_{\text{flip/conserve}}=1.230\text{e-}03$; $I_h=-4.142\text{e-}04$

virtual-photon powers: T 99.954%, L 0.046%

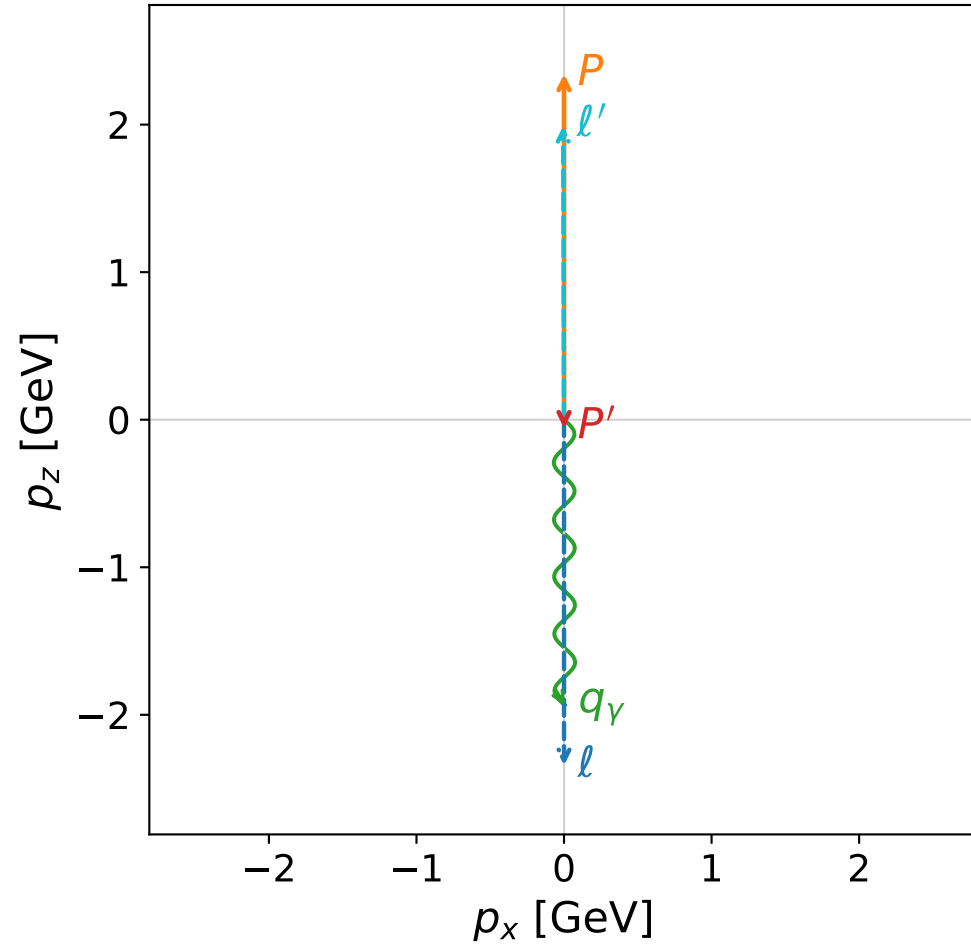
$R_L/T=4.624\text{e-}04$; $I_{TL}=-2.794\text{e-}03$

Leading normalized final-state amplitudes (N=2, retained=99.9%)

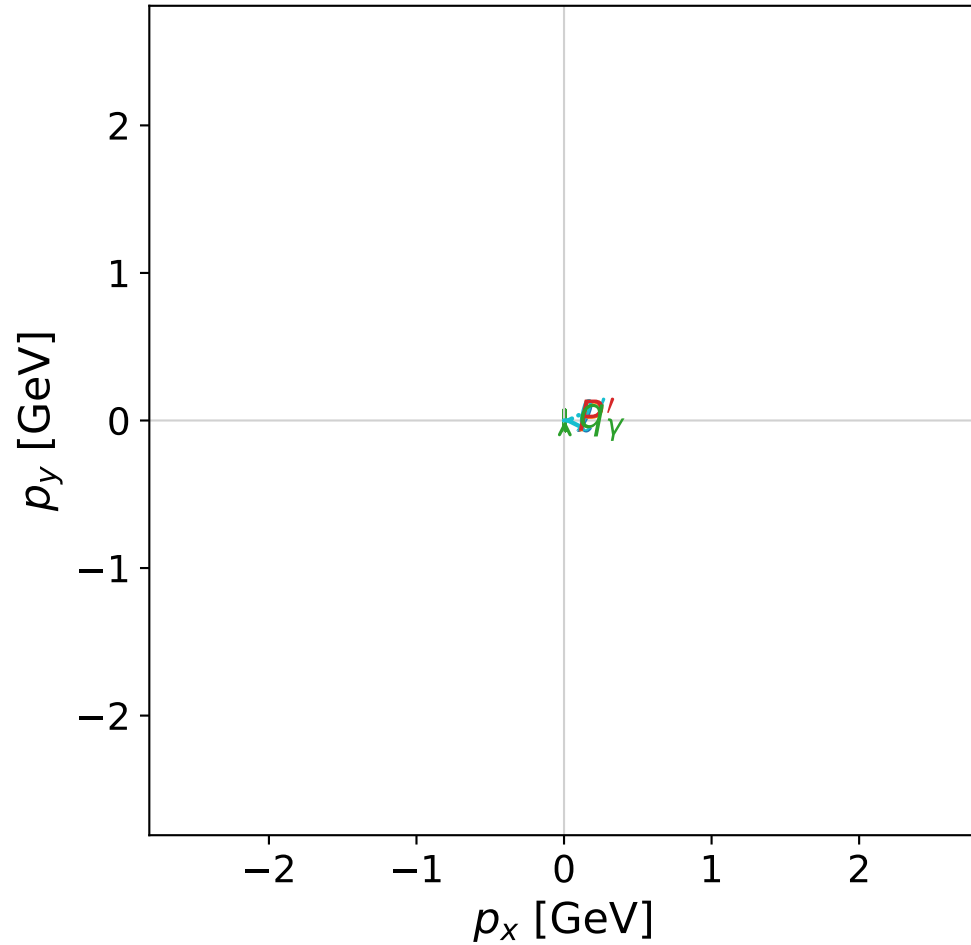


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



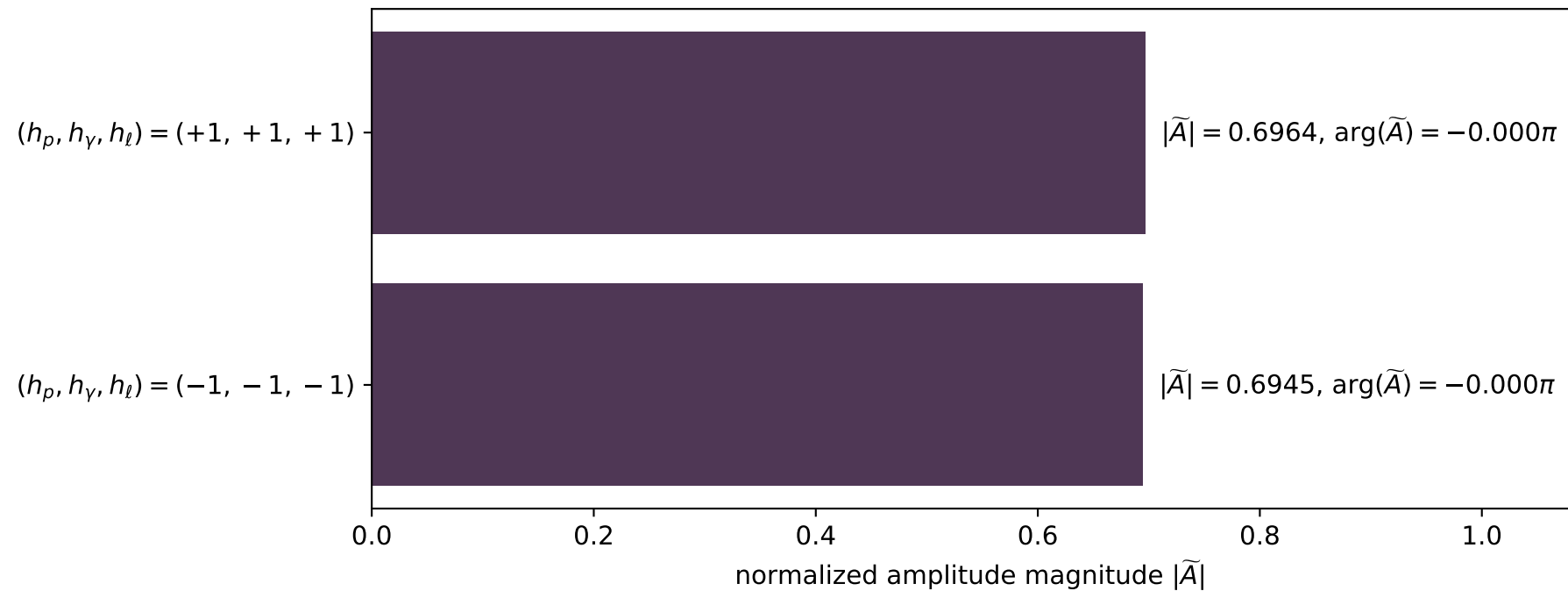
dghz_mixing_angles_polarization_04_local_minimum_245
 $d_{\text{GHZ}}=0.000119118$
 region: polarization_cluster_04_local_minimum_245
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0245
 lepton mixing angle: $\alpha_e=2.9733431$ rad
 incoming lepton: $-0.985879|+\rangle +0.167457|-\rangle$
 proton mixing angle: $\alpha_p=1.7395034$ rad
 incoming proton: $-0.167908|+\rangle +0.985803|-\rangle$

$s=23.6705$, $\sqrt{s}=4.86523$, $|\vec{P}|=2.3422$, $|\vec{P}'|=0.0567288$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.13863$, $\phi_{P'}=2.24178$, $\phi_\gamma=4.57737\text{e-}07$
 $E_\gamma=1.9344$, $\theta_{\ell'}=0.00287975$, $\phi_{\ell'}=3.14159$
 $Q^2=18.6544$, $x_B=0.845219$, $t=-3.24792$, $W^2=4.29593$, $y=0.968399$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3422$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3422$
 P $E= 2.5230$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3422$
 ℓ' $E= 1.9911$ $p_x= -0.0057$ $p_y= -0.0000$ $p_z= 1.9911$
 P' $E= 0.9397$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0567$
 q_γ $E= 1.9344$ $p_x= 0.0057$ $p_y= 0.0000$ $p_z= -1.9344$

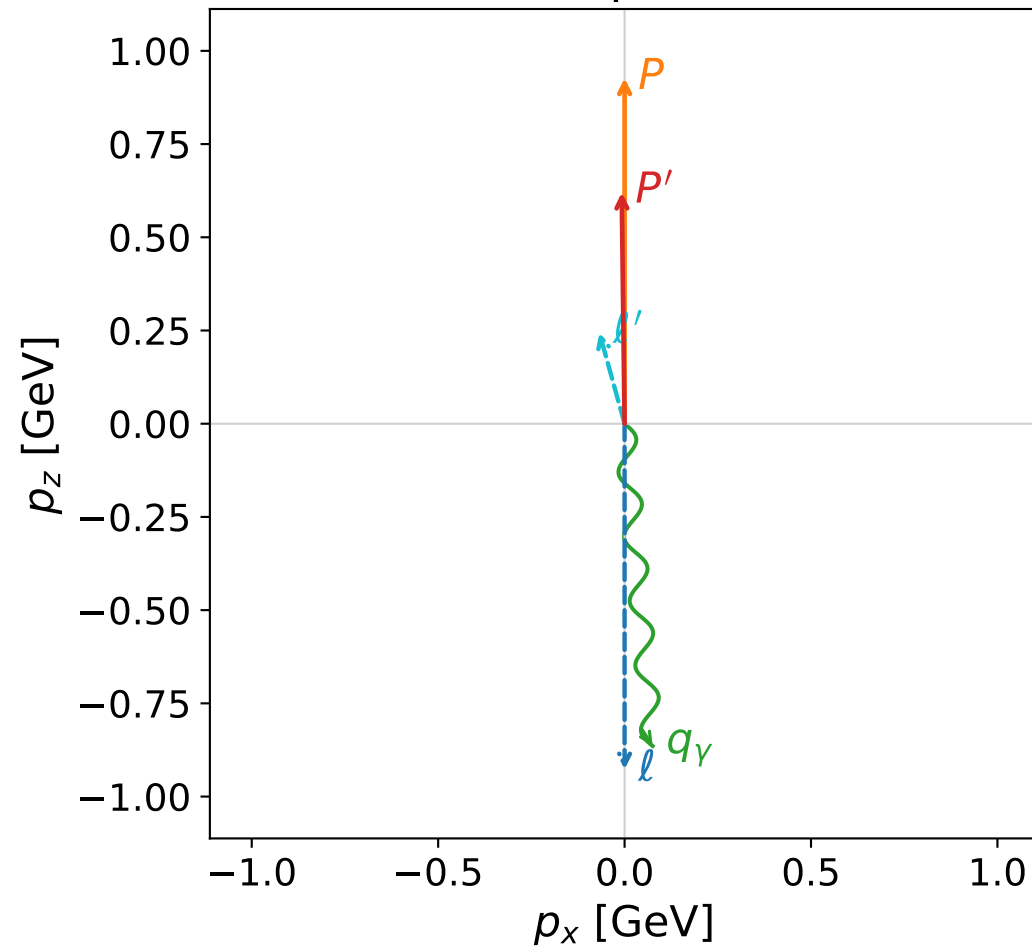
GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 5.841%, conserve 94.159%
 $R_{\text{flip/conserve}}=6.203\text{e-}02$; $I_h=-5.513\text{e-}02$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_L/T=2.847\text{e-}07$; $I_{TL}=-1.132\text{e-}04$

Leading normalized final-state amplitudes
 (N=2, retained=96.7%)

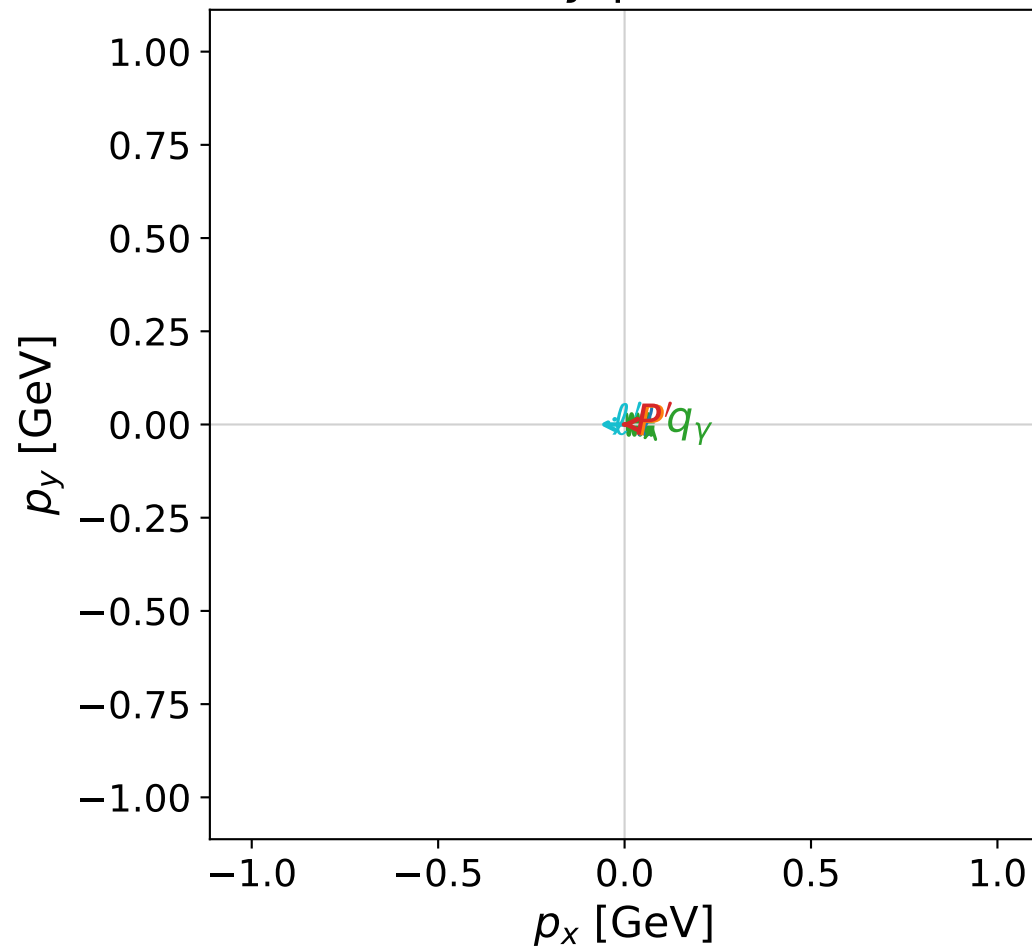


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

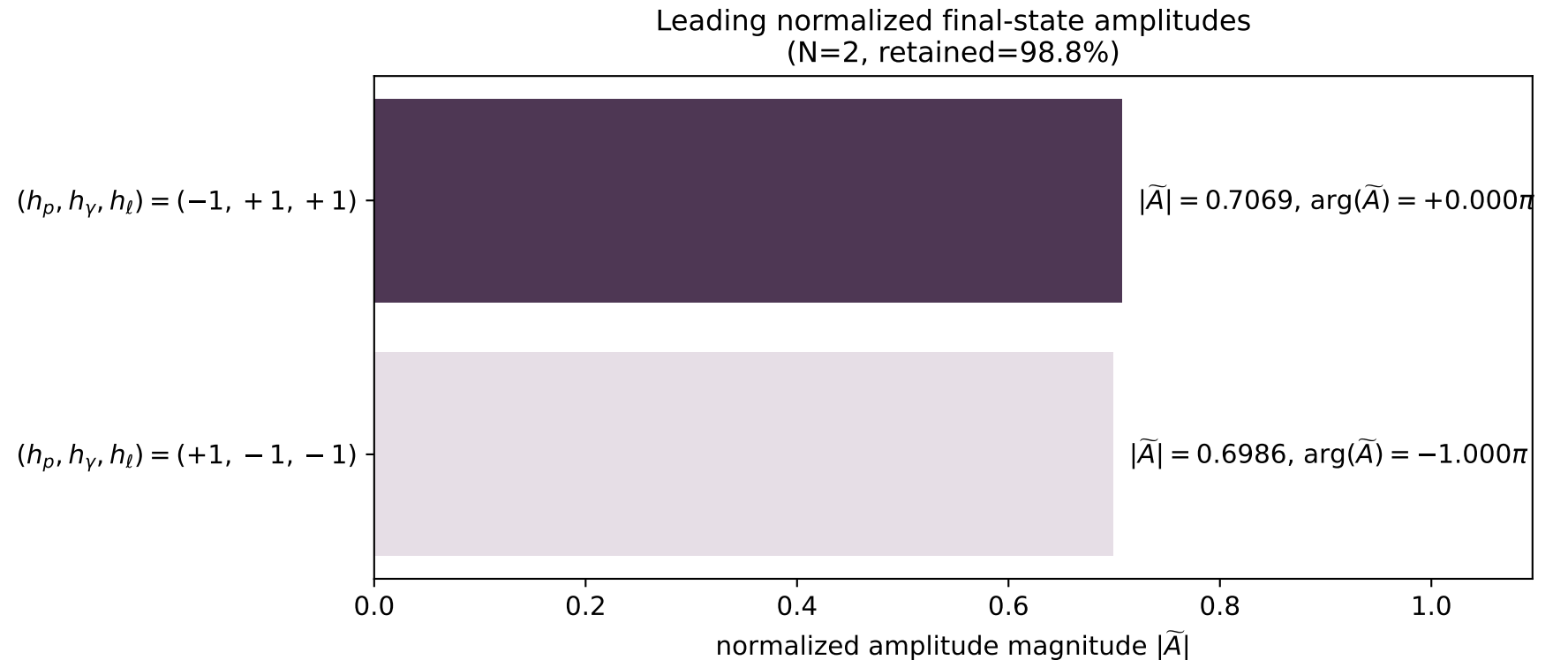


dghz_mixing_angles_polarization_04_local_minimum_251
 $d_{\text{GHZ}}=0.000133889$
 region: polarization_cluster_04_local_minimum_251
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0251
 lepton mixing angle: $\alpha_e=2.7320295$ rad
 incoming lepton: $-0.917295|+\rangle +0.398209|-\rangle$
 proton mixing angle: $\alpha_p=1.949413$ rad
 incoming proton: $-0.369635|+\rangle +0.929177|-\rangle$

$s=5.04265$, $\sqrt{s}=2.24559$, $|\vec{P}|=0.926887$, $|\vec{P}'|=0.620376$
 $\theta_{p'}=0.0123959$, $\theta_\gamma=3.05471$, $\phi_{p'}=3.14155$, $\phi_\gamma=6.28317$
 $E_\gamma=0.867702$, $\theta_{l'}=0.270165$, $\phi_{l'}=3.14158$
 $Q^2=0.922051$, $x_B=0.233591$, $t=-0.0563611$, $W^2=3.90509$, $y=0.948228$

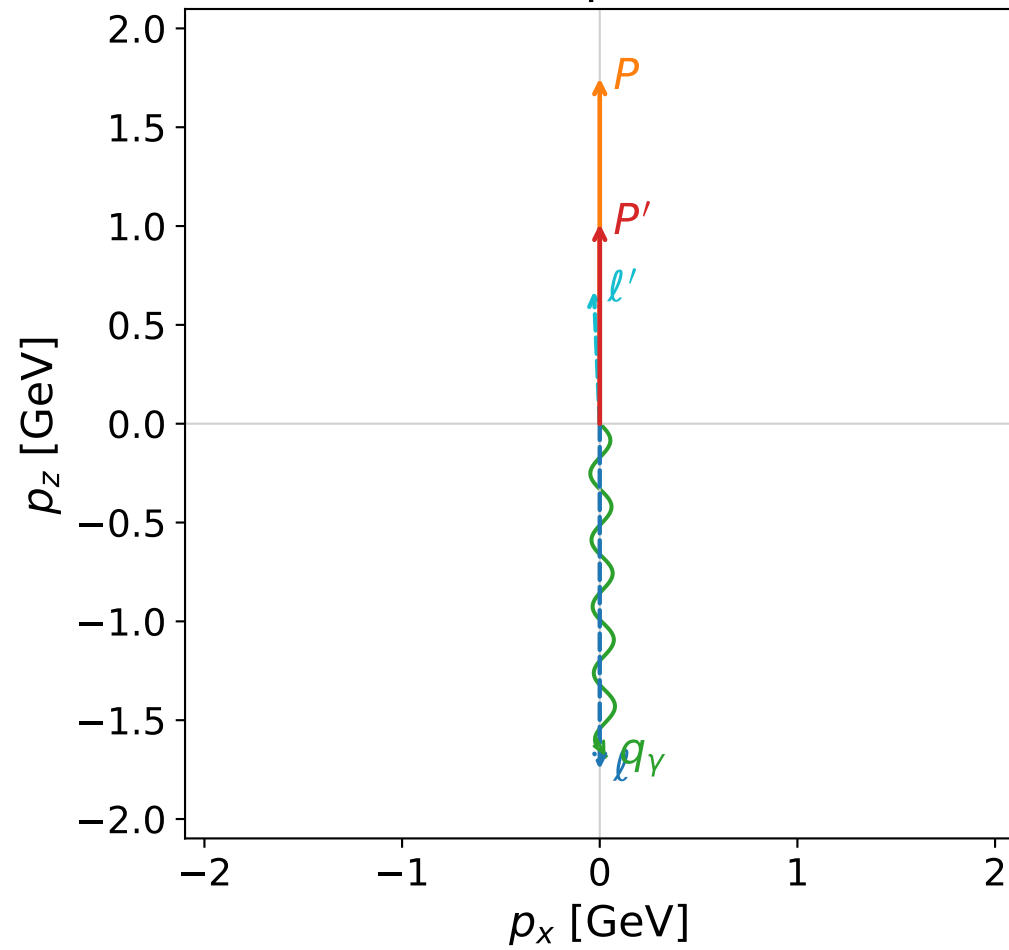
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.9269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9269$
 P $E= 1.3187$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9269$
 l' $E= 0.2533$ $p_x= -0.0076$ $p_y= 0.0000$ $p_z= 0.2441$
 P' $E= 1.1246$ $p_x= -0.0077$ $p_y= 0.0000$ $p_z= 0.6203$
 q_γ $E= 0.8677$ $p_x= 0.0753$ $p_y= -0.0000$ $p_z= -0.8644$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.009%, conserve 99.991%
 $R_{\text{flip}}/\text{conserve}=9.165\text{e-}05$; $I_h=-4.246\text{e-}04$
 virtual-photon powers: T 98.831%, L 1.169%
 $R_L/T=1.183\text{e-}02$; $I_{TL}=+2.749\text{e-}02$

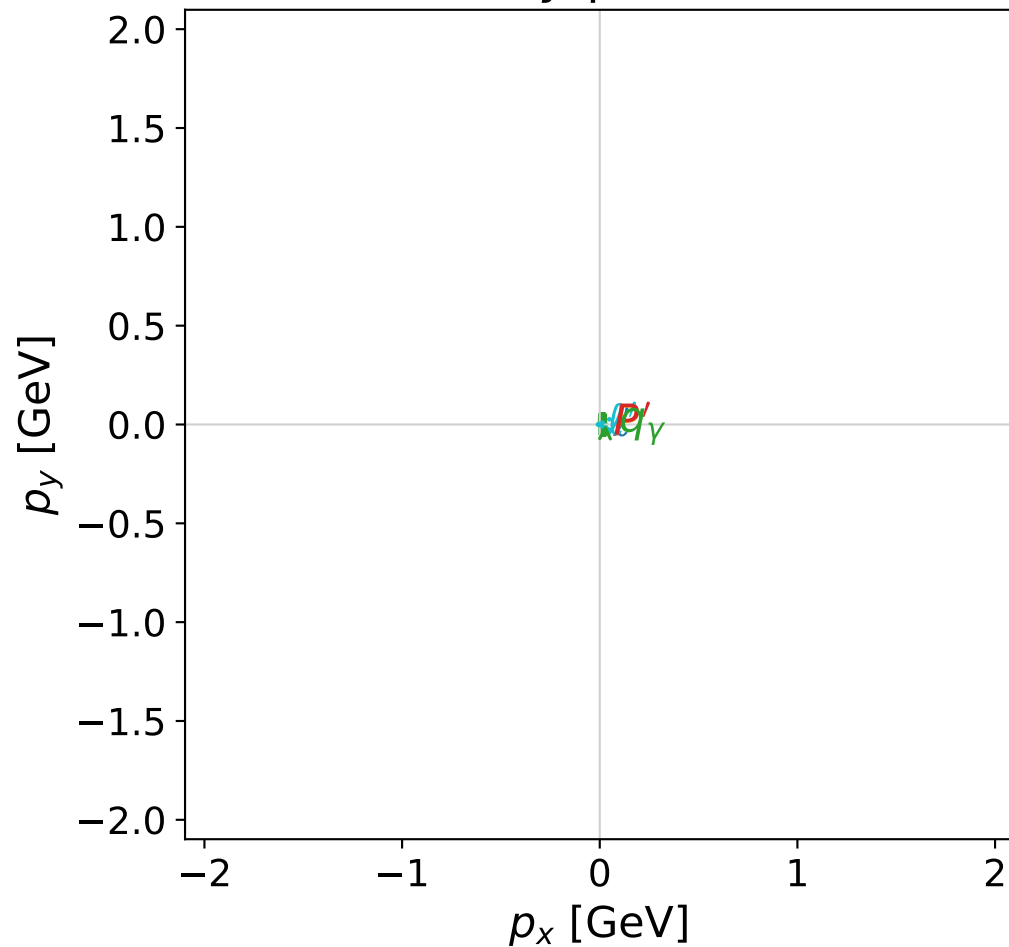


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

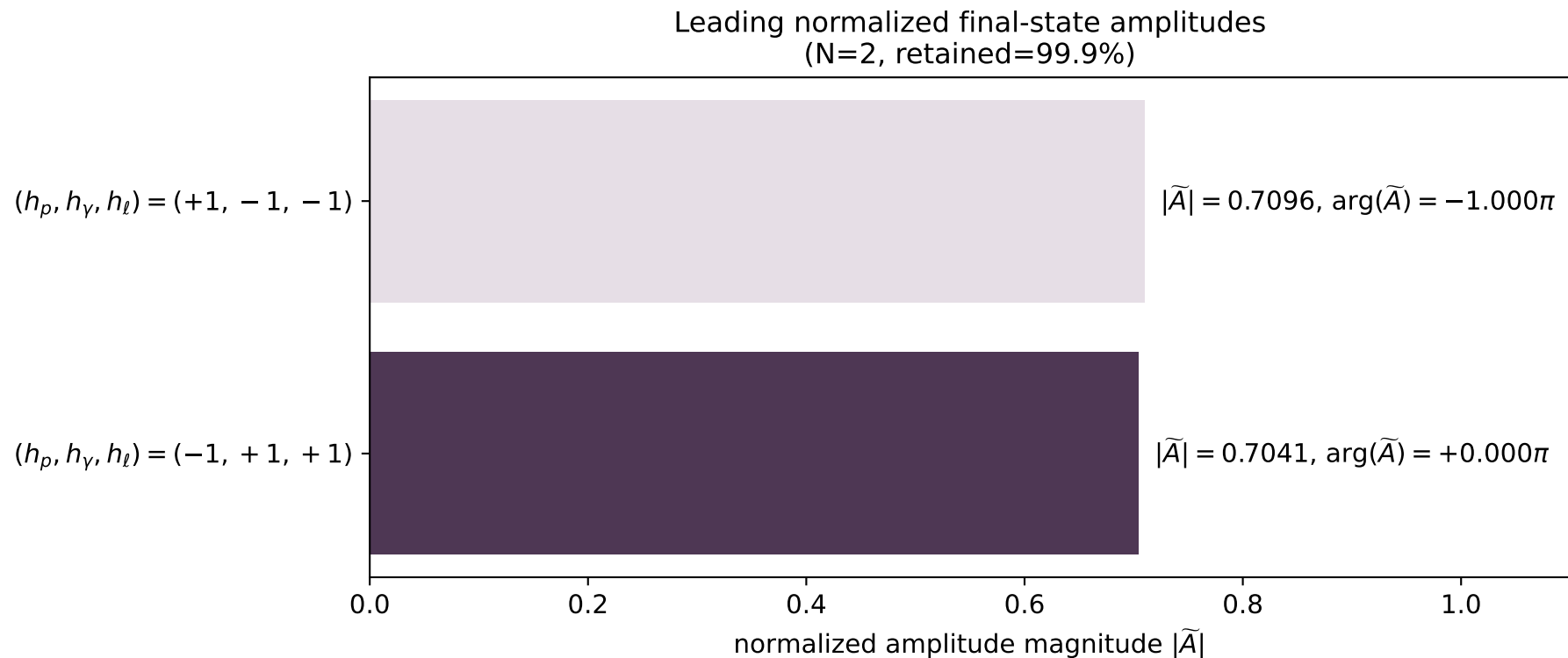


dghz_mixing_angles_polarization_04_local_minimum_252
 $d_{\text{GHZ}}=0.000137569$
 region: polarization_cluster_04_local_minimum_252
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0252
 lepton mixing angle: $\alpha_e=2.8965137$ rad
 incoming lepton: $-0.970118|+\rangle +0.242633|-\rangle$
 proton mixing angle: $\alpha_p=1.8140654$ rad
 incoming proton: $-0.240877|+\rangle +0.970556|-\rangle$

$s=13.9325$, $\sqrt{s}=3.73262$, $|\vec{P}|=1.74845$, $|\vec{P}'|=1.01096$
 $\theta_{P'}=0$, $\theta_\gamma=3.12412$, $\phi_{P'}=5.03716$, $\phi_\gamma=6.28317$
 $E_\gamma=1.68205$, $\theta_{l'}=0.0437779$, $\phi_{l'}=3.14158$
 $Q^2=4.69394$, $x_B=0.368621$, $t=-0.177771$, $W^2=8.9197$, $y=0.975575$

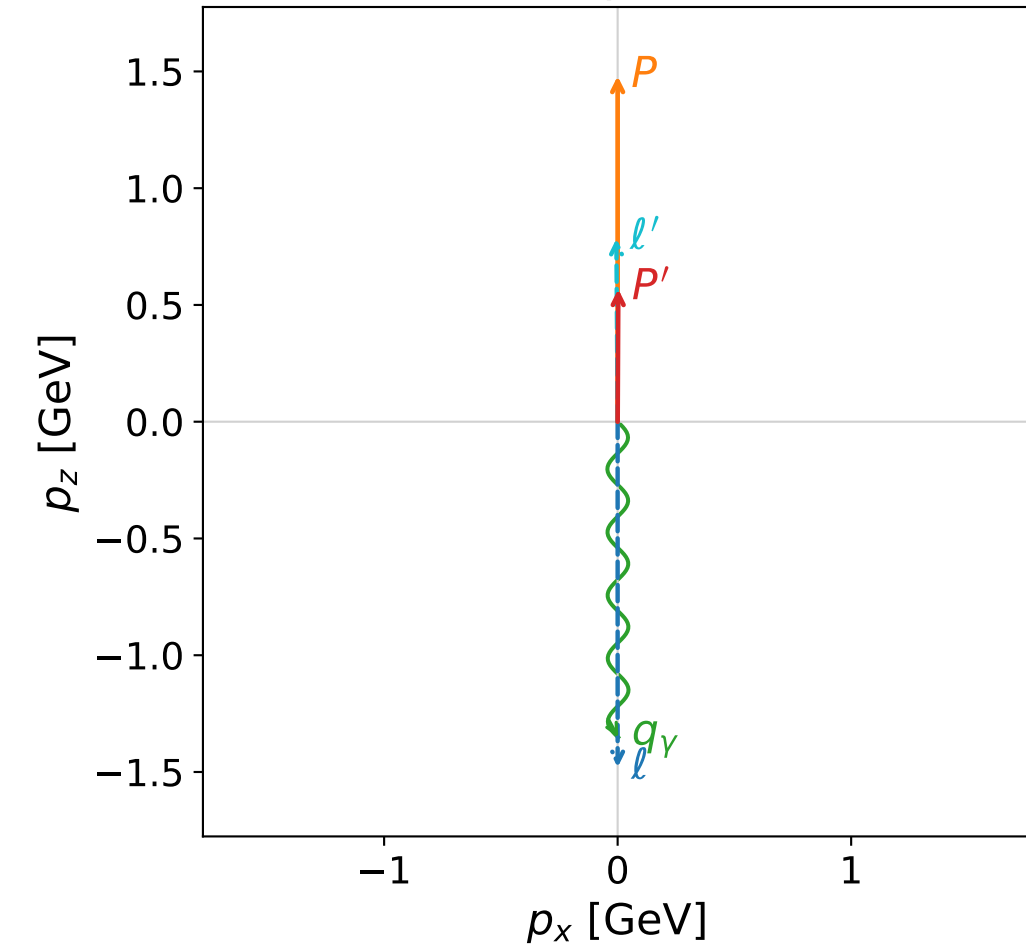
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.7485$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7485$
 P $E= 1.9842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7485$
 l' $E= 0.6715$ $p_x= -0.0294$ $p_y= 0.0000$ $p_z= 0.6708$
 P' $E= 1.3791$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.0110$
 q_γ $E= 1.6821$ $p_x= 0.0294$ $p_y= -0.0000$ $p_z= -1.6818$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.208%, conserve 99.792%
 $R_{\text{flip/conserve}}=2.083\text{e-}03$; $I_h=-2.263\text{e-}03$
 virtual-photon powers: T 99.984%, L 0.016%
 $R_L/T=1.565\text{e-}04$; $I_{TL}=-5.723\text{e-}04$

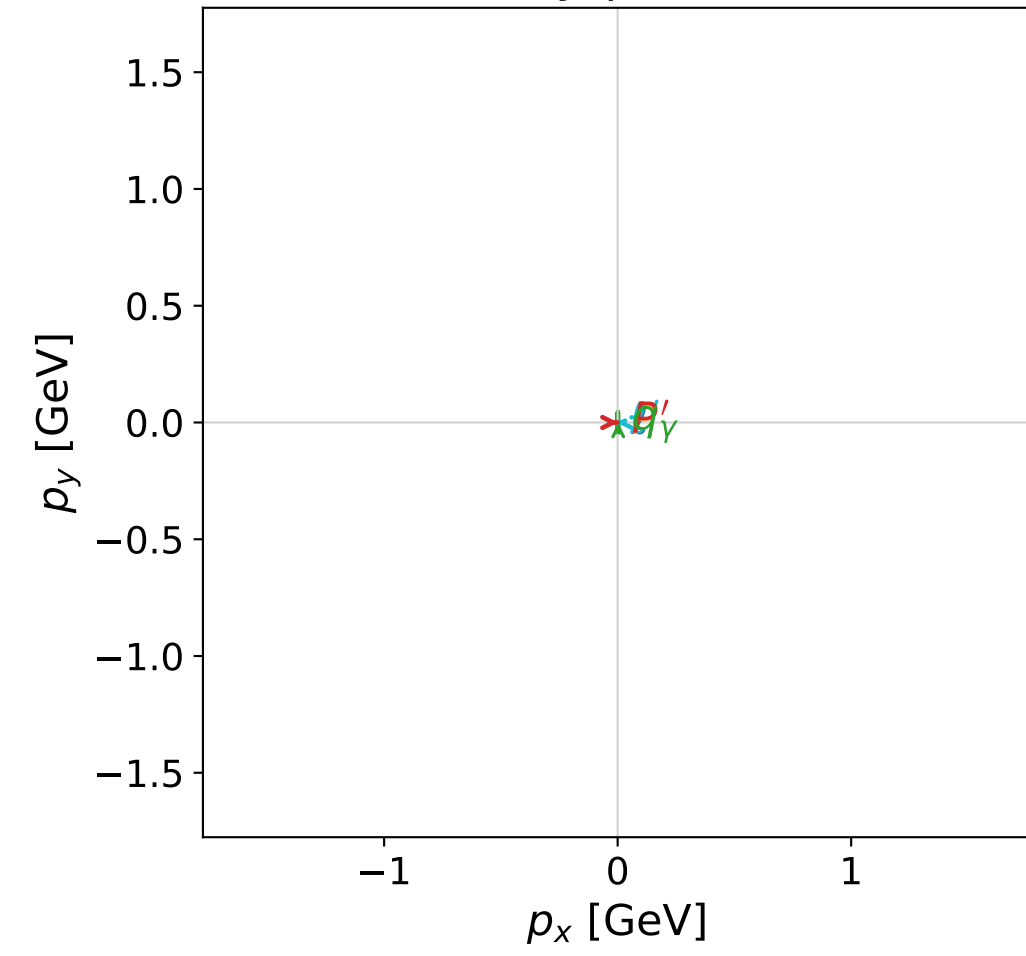


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

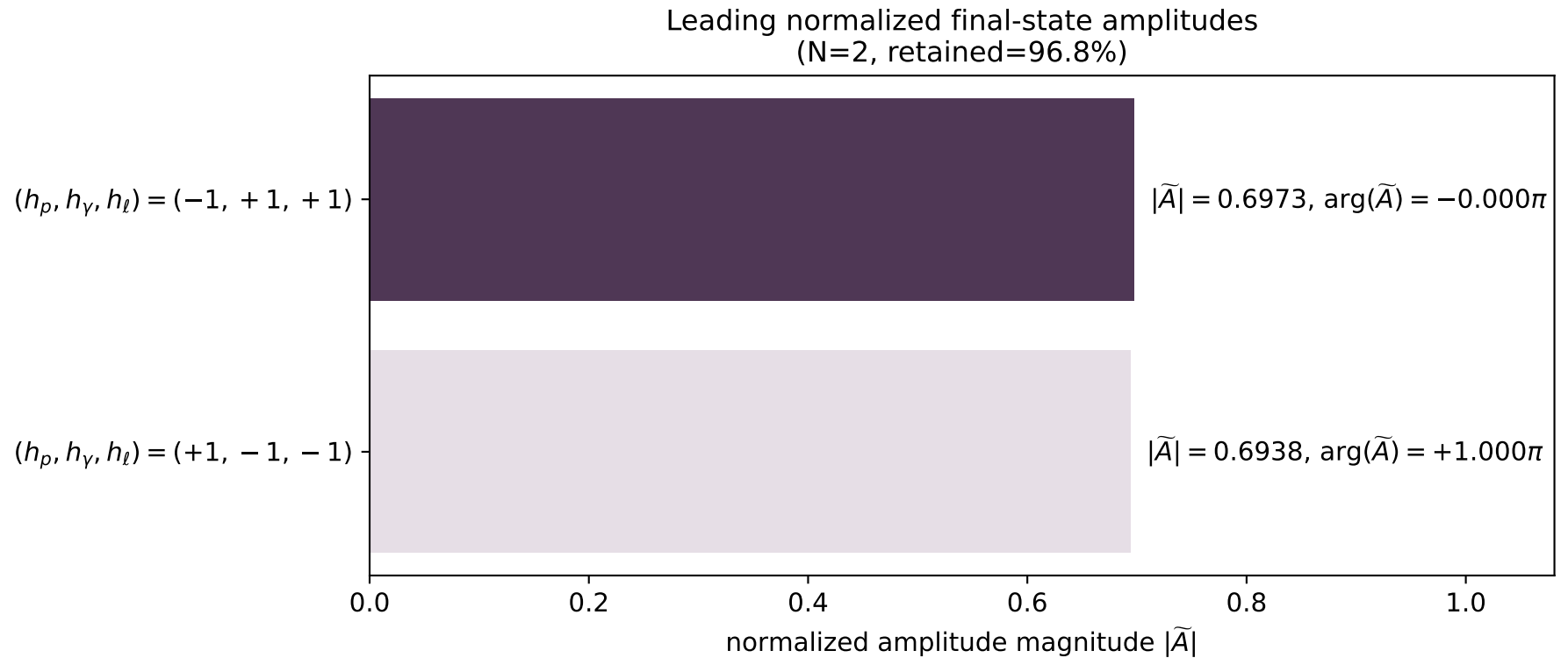


dghz_mixing_angles_polarization_04_local_minimum_293
 $d_{\text{GHZ}}=0.000295565$
 region: polarization_cluster_04_local_minimum_293
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0293
 lepton mixing angle: $\alpha_e=2.6023939$ rad
 incoming lepton: $-0.85812|+\rangle +0.513449|-\rangle$
 proton mixing angle: $\alpha_p=2.1130396$ rad
 incoming proton: $-0.516059|+\rangle +0.856553|-\rangle$

$s=10.4463$, $\sqrt{s}=3.23207$, $|\vec{P}|=1.47992$, $|\vec{P}'|=0.568077$
 $\theta_{p'}=0.00483204$, $\theta_\gamma=3.13964$, $\phi_{p'}=6.28121$, $\phi_\gamma=0$
 $E_\gamma=1.35176$, $\theta_{l'}=0.00687461$, $\phi_{l'}=3.14059$
 $Q^2=4.63923$, $x_B=0.507591$, $t=-0.401758$, $W^2=5.38032$, $y=0.955392$

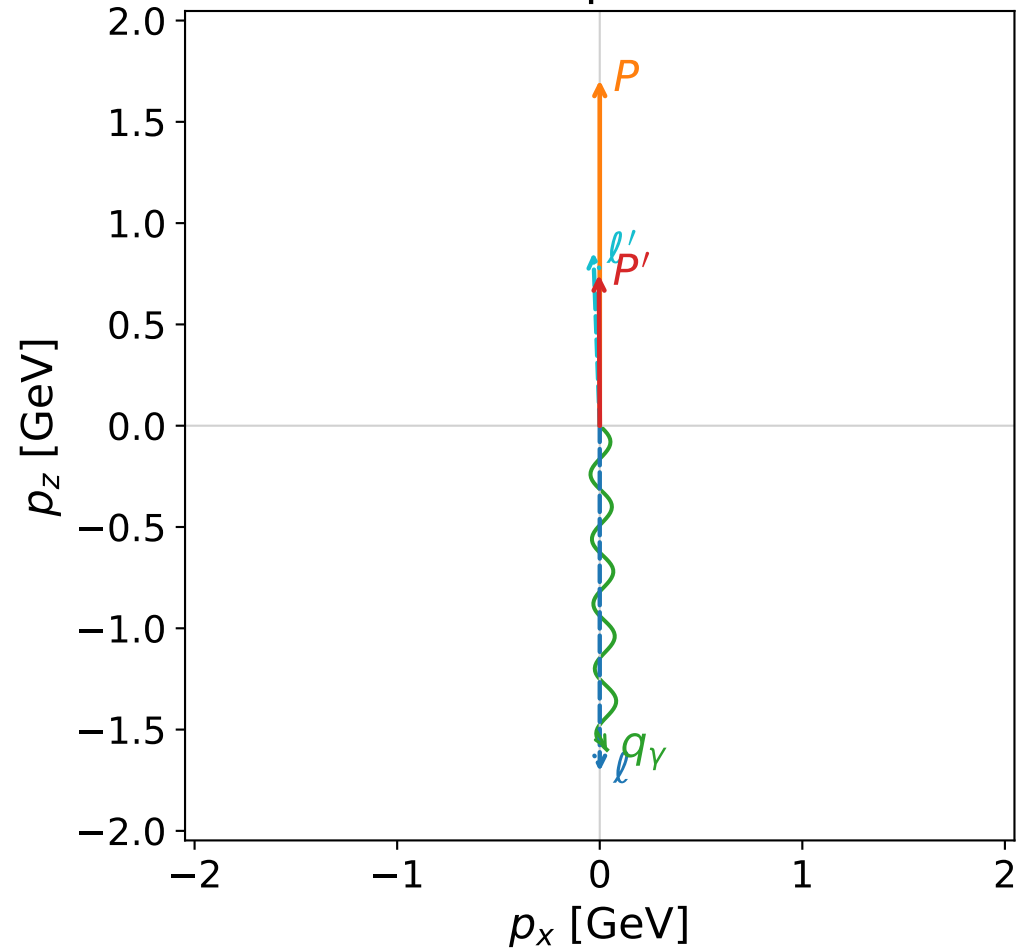
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.4799$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4799$
 P $E= 1.7521$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.4799$
 l' $E= 0.7837$ $p_x= -0.0054$ $p_y= 0.0000$ $p_z= 0.7837$
 P' $E= 1.0966$ $p_x= 0.0027$ $p_y= -0.0000$ $p_z= 0.5681$
 q_γ $E= 1.3518$ $p_x= 0.0026$ $p_y= 0.0000$ $p_z= -1.3518$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 3.894%, conserve 96.106%
 $R_{\text{flip/conserve}}=4.052\text{e-}02$; $I_h=-1.410\text{e-}02$
 virtual-photon powers: T 100.000%, L 0.000%
 $R_{L/T}=1.385\text{e-}06$; $I_{TL}=-6.576\text{e-}04$

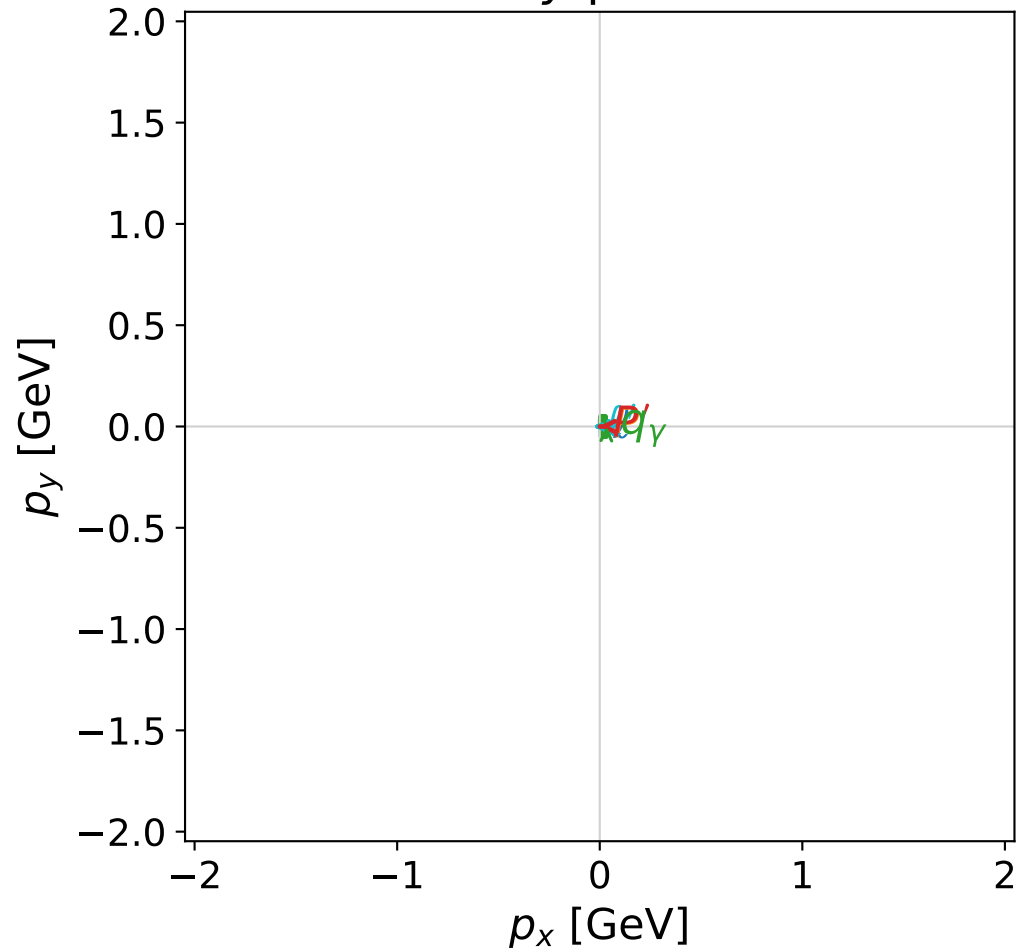


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

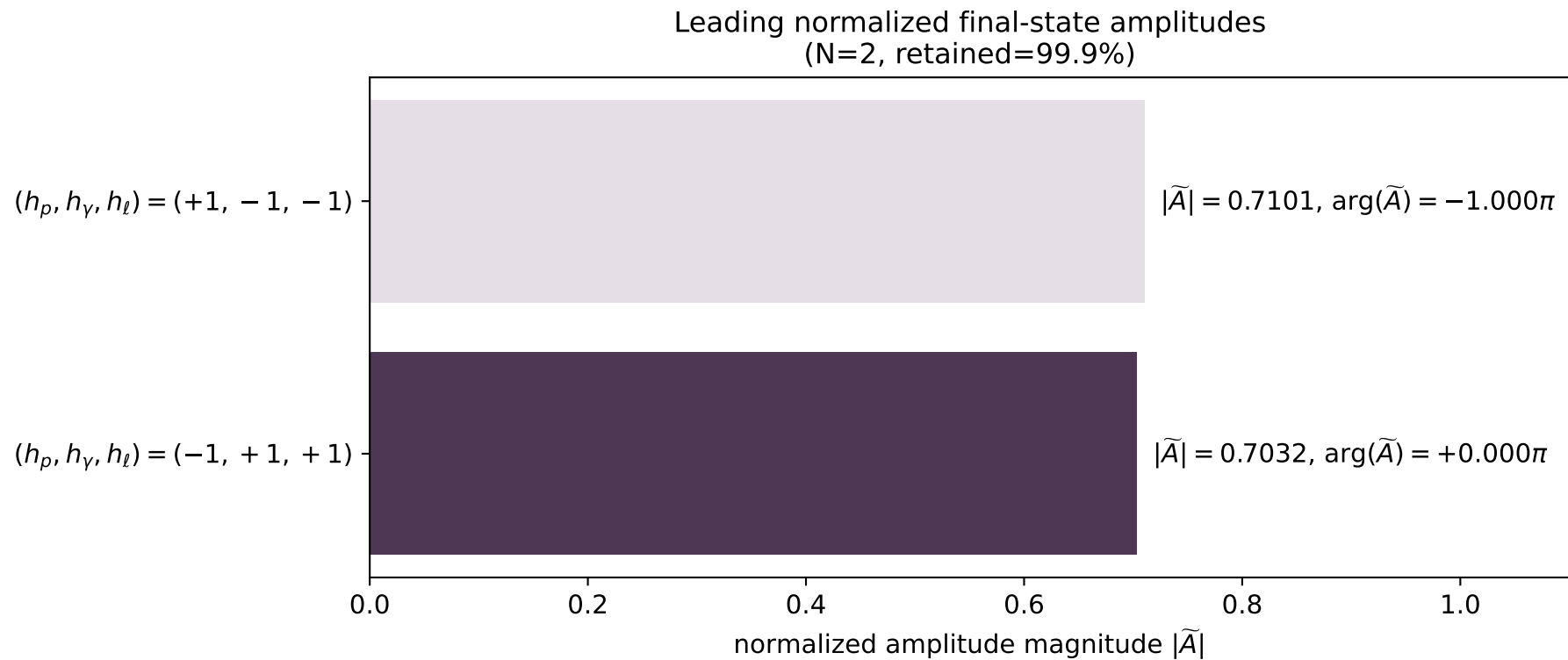


dghz_mixing_angles_polarization_04_local_minimum_297
 $d_{\text{GHZ}}=0.000306617$
 region: polarization_cluster_04_local_minimum_297
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0297
 lepton mixing angle: $\alpha_e=3.0107596$ rad
 incoming lepton: $-0.991454|+\rangle + 0.13046|-\rangle$
 proton mixing angle: $\alpha_p=1.6989407$ rad
 incoming proton: $-0.127794|+\rangle + 0.991801|-\rangle$

$s=13.3426$, $\sqrt{s}=3.65276$, $|\vec{P}|=1.70594$, $|\vec{P}'|=0.745662$
 $\theta_{P'}=0.00520095$, $\theta_Y=3.11943$, $\phi_{P'}=3.14107$, $\phi_Y=0$
 $E_Y=1.59997$, $\theta_{\ell'}=0.0369583$, $\phi_{\ell'}=3.14166$
 $Q^2=5.82901$, $x_B=0.483769$, $t=-0.361859$, $W^2=7.09999$, $y=0.96681$

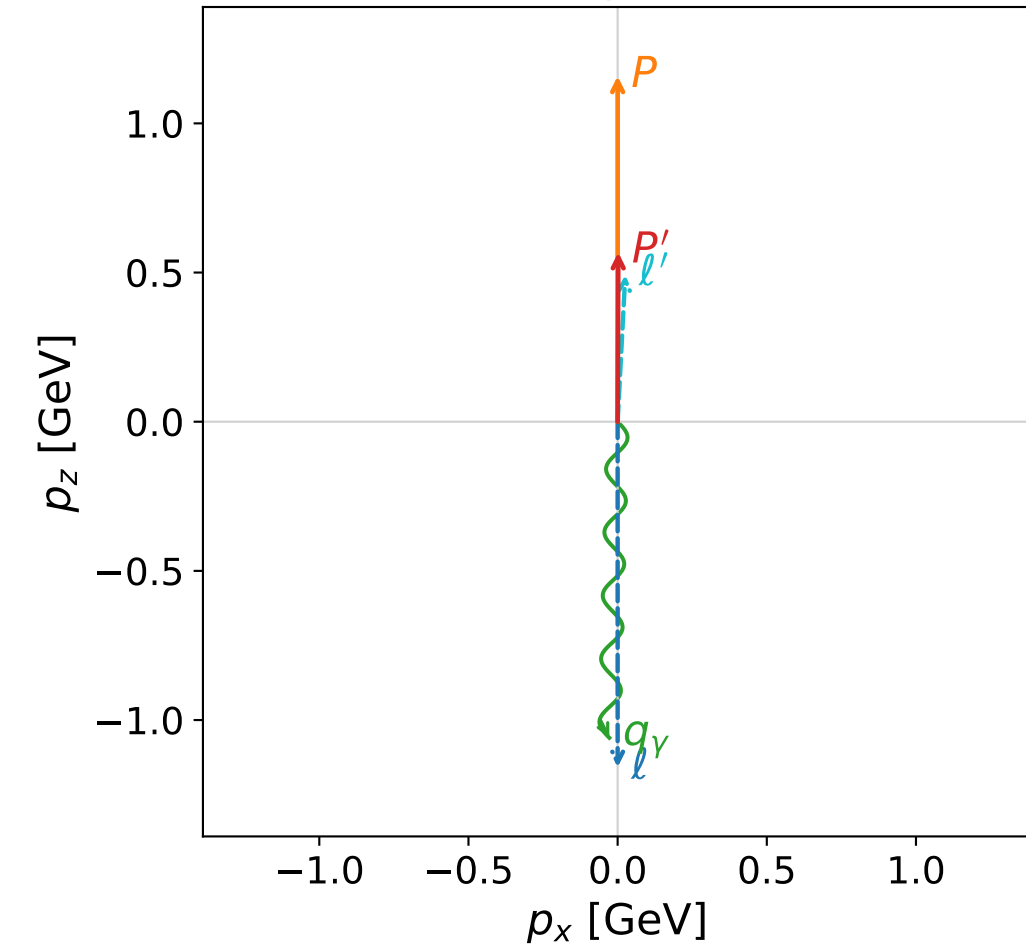
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7059$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7059$
 P $E= 1.9468$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7059$
 ℓ' $E= 0.8545$ $p_x= -0.0316$ $p_y= -0.0000$ $p_z= 0.8539$
 P' $E= 1.1983$ $p_x= -0.0039$ $p_y= 0.0000$ $p_z= 0.7457$
 q_Y $E= 1.6000$ $p_x= 0.0355$ $p_y= 0.0000$ $p_z= -1.5996$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.464%, conserve 99.536%
 $R_{\text{flip/conserve}}=4.665\text{e-}03$; $I_h=-5.839\text{e-}03$
 virtual-photon powers: T 99.953%, L 0.047%
 $R_L/T=4.706\text{e-}04$; $I_{TL}=+3.914\text{e-}03$

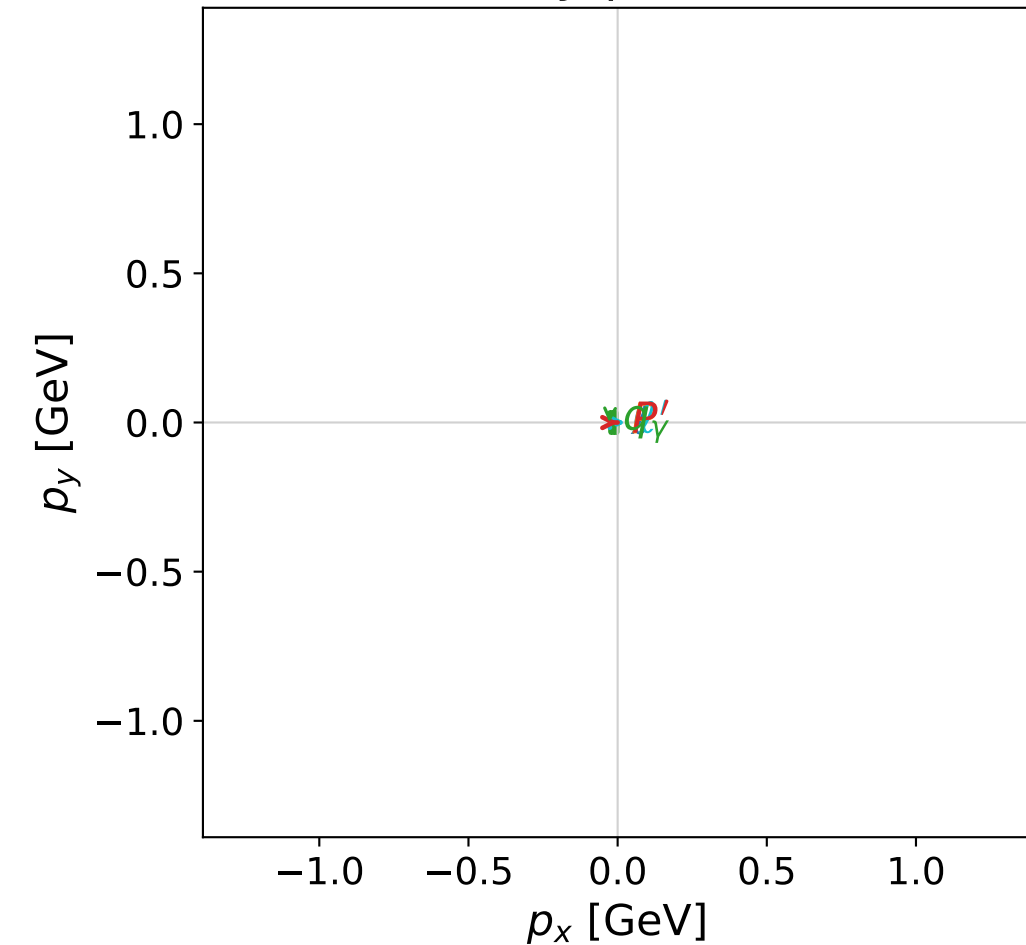


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

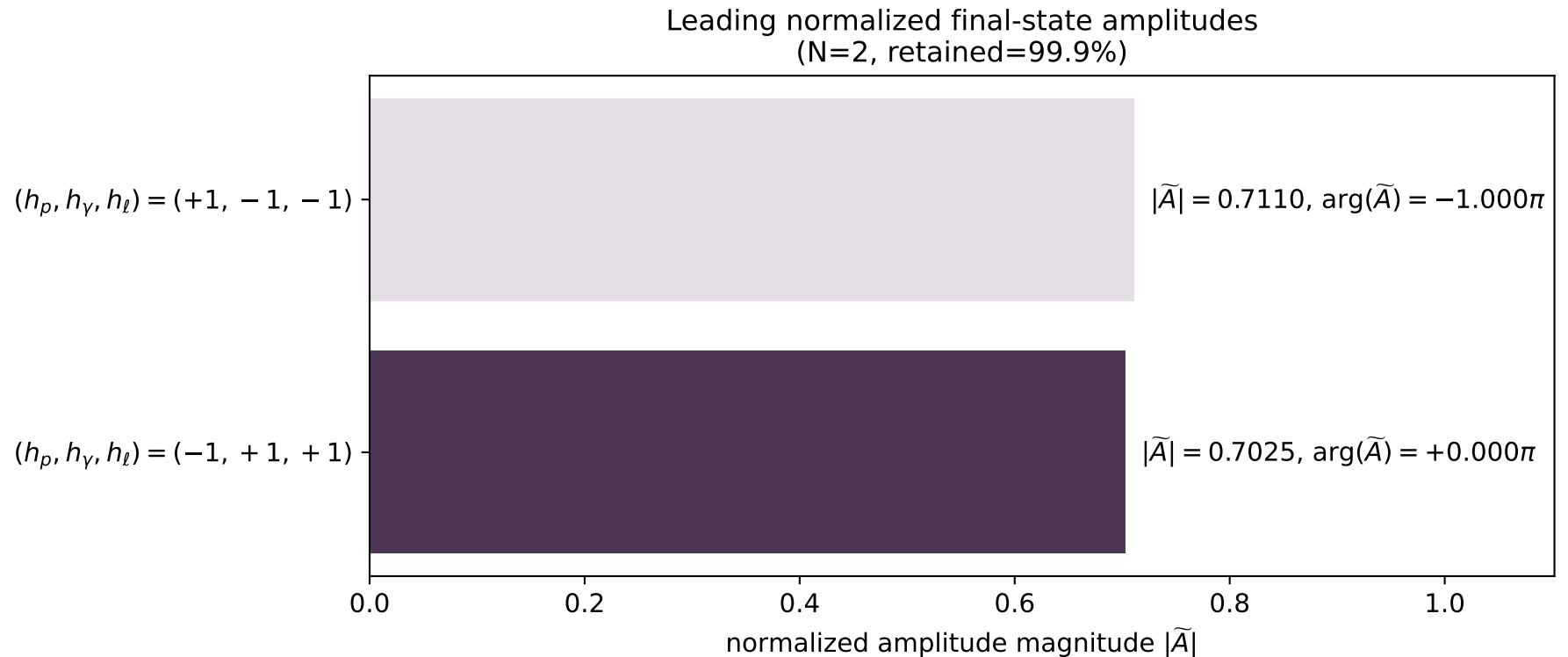


dghz_mixing_angles_polarization_04_local_minimum_319
 $d_{\text{GHZ}}=0.000411475$
 region: polarization_cluster_04_local_minimum_319
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0319
 lepton mixing angle: $\alpha_e=1.704901$ rad
 incoming lepton: $-0.133703|+\rangle +0.991021|-\rangle$
 proton mixing angle: $\alpha_p=3.0080191$ rad
 incoming proton: $-0.991092|+\rangle +0.133177|-\rangle$

$s=7.01694$, $\sqrt{s}=2.64895$, $|\vec{P}|=1.1584$, $|\vec{P}'|=0.567777$
 $\theta_{p'}=0.00378308$, $\theta_\gamma=3.11539$, $\phi_{p'}=0.00339324$, $\phi_\gamma=3.14154$
 $E_\gamma=1.05998$, $\theta_{l'}=0.0520539$, $\phi_{l'}=6.28284$
 $Q^2=2.28056$, $x_B=0.392633$, $t=-0.193536$, $W^2=4.40765$, $y=0.946437$

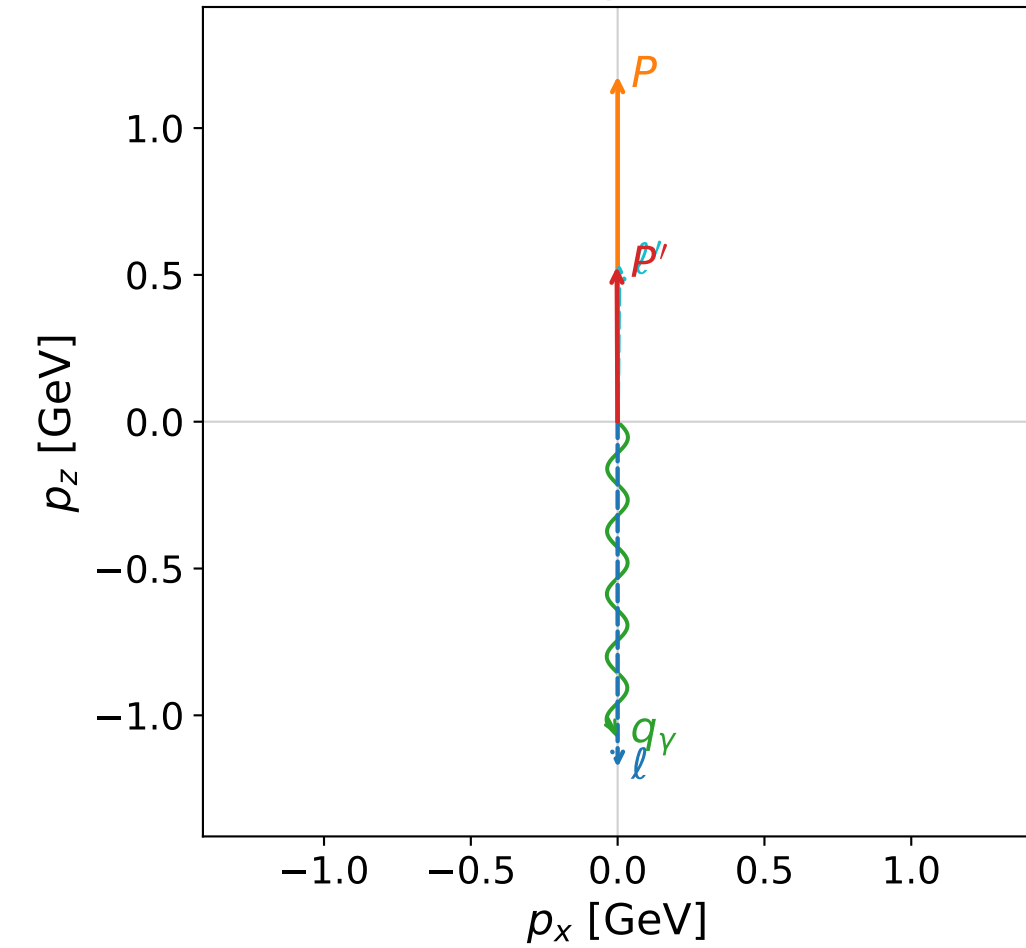
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.1584$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1584$
 P $E= 1.4905$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1584$
 l' $E= 0.4925$ $p_x= 0.0256$ $p_y= -0.0000$ $p_z= 0.4918$
 P' $E= 1.0965$ $p_x= 0.0021$ $p_y= 0.0000$ $p_z= 0.5678$
 q_γ $E= 1.0600$ $p_x= -0.0278$ $p_y= 0.0000$ $p_z= -1.0596$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 0.635%, conserve 99.365%
 $R_{\text{flip/conserve}}=6.386\text{e-}03$; $I_h=-9.149\text{e-}03$
 virtual-photon powers: T 99.817%, L 0.183%
 $R_L/T=1.833\text{e-}03$; $I_{TL}=+4.055\text{e-}03$

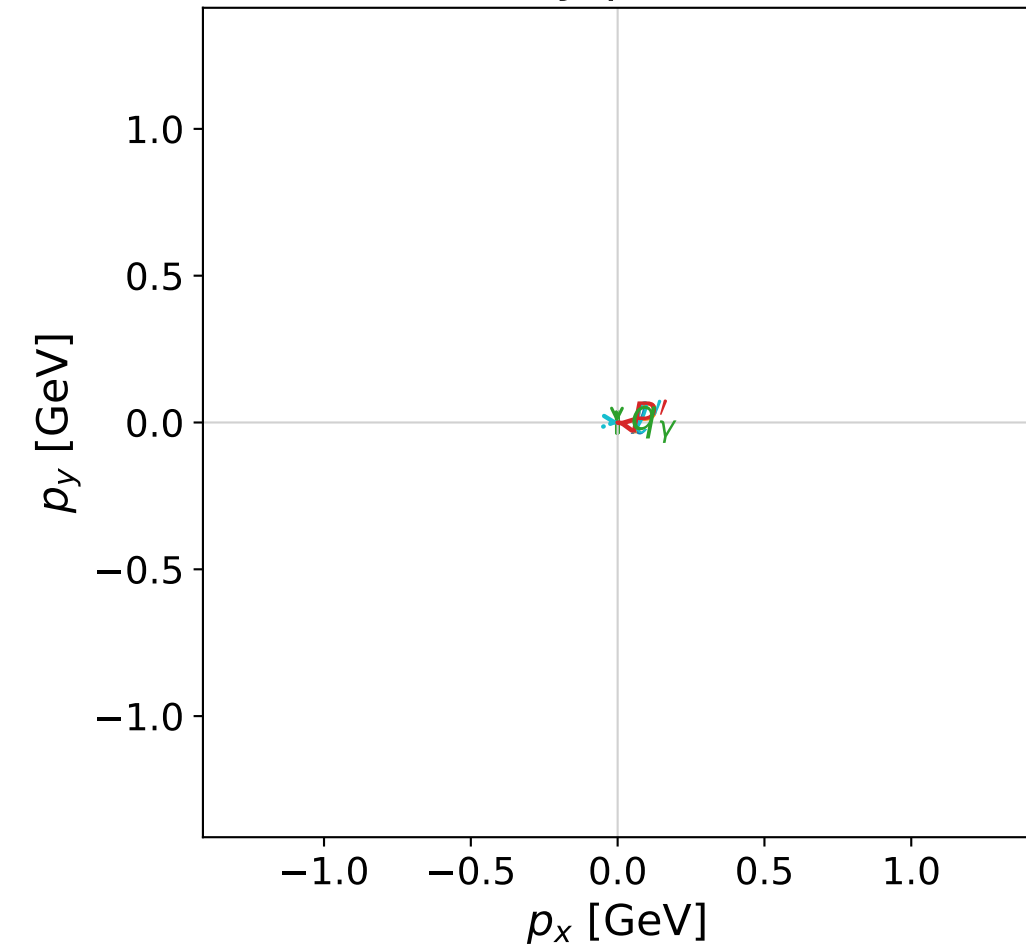


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_04_local_minimum_336
 $d_{\text{GHZ}}=0.000558565$
 region: polarization_cluster_04_local_minimum_336
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0336
 lepton mixing angle: $\alpha_e=2.0949563$ rad
 incoming lepton: $-0.500486|+\rangle +0.865745|-\rangle$
 proton mixing angle: $\alpha_p=2.6121434$ rad
 incoming proton: $-0.863085|+\rangle +0.505058|-\rangle$

$s=7.19362$, $\sqrt{s}=2.68209$, $|\vec{P}|=1.17702$, $|\vec{P}'|=0.528084$
 $\theta_{p'}=0.00510742$, $\theta_\gamma=3.13917$, $\phi_{p'}=2.97399$, $\phi_\gamma=3.14159$
 $E_\gamma=1.06685$, $\theta_{l'}=0.00977172$, $\phi_{l'}=6.19762$
 $Q^2=2.53666$, $x_B=0.425599$, $t=-0.237414$, $W^2=4.30339$, $y=0.944002$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=1.1770$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-1.1770$
 P $E=1.5051$ $p_x=0.0000$ $p_y=0.0000$ $p_z=1.1770$
 ℓ' $E=0.5388$ $p_x=0.0052$ $p_y=-0.0004$ $p_z=0.5388$
 P' $E=1.0764$ $p_x=-0.0027$ $p_y=0.0004$ $p_z=0.5281$
 q_γ $E=1.0669$ $p_x=-0.0026$ $p_y=-0.0000$ $p_z=-1.0669$

GHZ mechanism: helicity conserving (full H G C)
 helicity powers: flip 4.135%, conserve 95.865%
 $R_{\text{flip/conserve}}=4.313\text{e-}02$; $I_h=-1.628\text{e-}02$
 virtual-photon powers: T 99.999%, L 0.001%
 $R_{L/T}=5.755\text{e-}06$; $I_{TL}=-1.436\text{e-}03$

